

Privacy-preserving and secure spectrum sharing for database-driven cognitive radio networks

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ABSTRACT

Database-driven cognitive radio networks (DB-CRNs) enable dynamic spectrum sharing through geolocation databases but introduce critical security and privacy challenges, including mandatory location disclosure, susceptibility to location spoofing, and denial-of-service (DoS) attacks on centralized services. Existing approaches address these issues in isolation and lack a unified, regulation-compliant solution under realistic adversarial conditions. In this work, we present a unified security framework for DB-CRNs that simultaneously provides location privacy, user anonymity, verifiable location, and DoS resilience. Our framework, denoted as SLAPX, enables privacy-preserving spectrum queries using delegatable anonymous credentials, supports adaptive location assurance without revealing precise user location, and mitigates DoS attacks through verifiable delay functions (VDFs) combined with RLRS-based rate limiting. Extensive cryptographic benchmarking and network simulations demonstrate that SLAPX achieves significantly lower latency and communication overhead than existing solutions while effectively resisting location spoofing and DoS attacks. These results show that SLAPX is practical and well-suited for secure next-generation DB-CRN deployments.

1. Introduction

The rapid growth of wireless technologies, driven by mobile devices, Internet-of-Things (IoT) deployments, and emerging spectrum-hungry applications, has intensified pressure on traditionally static spectrum allocation policies [1–3]. To address spectrum underutilization, database-driven cognitive radio networks (DB-CRNs) and dynamic spectrum sharing have been introduced, enabling Secondary Users (SUs) to opportunistically access licensed spectrum owned by Primary Users (PUs) through regulatory-approved geolocation databases (DBs). A prominent example is the U.S. Citizens Broadband Radio Service (CBRS), which operates in the 3.5 GHz band originally allocated to federal and satellite services [1,4]. Moreover, dynamic spectrum sharing is expected to be a foundational element in future networks such as 6G systems [5].

While this paradigm significantly improves spectrum efficiency, it also introduces a wide range of security and privacy challenges. In particular, regulatory authorities (e.g., FCC and ITU) require users to continuously report sensitive operational information, including precise location, device identity, and transmission parameters, to centralized DBs. This requirement raises serious concerns regarding user privacy, anonymity, and long-term profiling [6,7]. Moreover, the location-centric design of DB-CRNs makes them inherently vulnerable to location spoofing attacks, in which malicious or compromised users falsify

their positions to obtain unauthorized spectrum access [8]. At the same time, the centralized reliance on spectrum databases, the broadcast nature of wireless communication, and the proliferation of low-cost devices further expose DB-CRNs to denial-of-service (DoS) attacks that can disrupt spectrum coordination and degrade network availability [9].

1.1. Open problems and design challenges

Despite substantial progress, secure DB-CRN design still lacks a unified and regulation-compliant solution that jointly supports anonymous spectrum access, strong location privacy, trustworthy location assurance, and resilience to DoS attacks under realistic adversarial conditions. The main technical challenges can be summarized as follows:

(i) *Location Privacy and Anonymity*: In DB-CRNs, regulatory requirements force unlicensed users (i.e., SUs) to disclose precise location, device characteristics, and spectrum requests to geolocation databases in order to obtain authorized channels and transmission parameters [1]. Although necessary for incumbent protection, auditability, and regulatory enforcement, this disclosure fundamentally conflicts with privacy-by-design, since users must trade privacy for spectrum access even at the query stage. Repeated queries can therefore be aggregated and

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correlated over time to infer identities, mobility patterns, and long-term behavioral profiles, enabling tracking and profiling attacks [4]. These risks persist even in the absence of external adversaries, as legitimate entities can still observe sensitive query metadata despite standard transport protections [10]. Moreover, location privacy without anonymity remains weak, while anonymity without unlinkability is still insufficient, since stable identifiers or repeated anonymous disclosures can enable session linkage and tracking [11]. Hence, DB-CRNs require lightweight and scalable mechanisms that jointly provide location privacy, anonymity, and unlinkability while remaining regulation-compliant and practical under realistic deployment constraints.

(ii) *Location Proof and Spoofing Resistance*: DB-CRNs are inherently location-based services in which spectrum availability decisions rely on real-time, user-reported location information stored in geolocation databases. Therefore, the accuracy and integrity of reported location data are critical for interference protection and fair spectrum allocation. Spoofed or falsified locations undermine regulatory authorization and can enable unauthorized spectrum access, causing interference, service disruption, and economic loss [12]. At the same time, directly revealing the precise location to prove correctness undermines privacy, whereas hiding it without additional safeguards weakens access control. This creates a fundamental tension between privacy and trustworthy spectrum authorization, especially under realistic settings with malicious or compromised users. As such, DB-CRNs need a practical, privacy-preserving, and proximity-backed location-assurance mechanism that resists spoofing and distance-fraud-style attacks without requiring fine-grained location disclosure. Detailed scheme-level limitations are discussed in Section 6.

(iii) *DoS Countermeasures and Spectrum Management*: In DB-CRNs, centralized geolocation databases interact with large numbers of users and IoT devices, making them high-value choke points whose unavailability can disrupt spectrum access at scale [13]. This risk is exacerbated by cost asymmetry: adversaries can issue many low-cost requests, while the infrastructure bears substantially higher verification and processing overhead. Thus, even weak attackers can flood spectrum databases with illegitimate queries, degrading spectrum coordination, reducing service availability, and impairing responsiveness during spectrum access and related operations. The core challenge is thus not only detecting abuse but also embedding lightweight and proactive request-throttling into the spectrum-access workflow itself, without shifting the attack surface or imposing excessive overhead on legitimate users. Scheme-level discussion of existing defenses is deferred to Section 6.

1.2. Our contributions

We develop and evaluate an efficient framework, namely SLAPX, that leverages advanced cryptographic primitives to jointly address privacy-preserving spectrum access, proximity-backed location assurance, and DoS resilience in DB-CRNs under realistic adversarial settings. This paper substantially extends our preliminary SLAP framework¹ with the following key capabilities:

- **Efficient location privacy-preserving and anonymous spectrum query**: SLAPX provides anonymity, location privacy, and unlinkability throughout proof-of-location (PoL) acquisition, spectrum querying, and service requests while remaining compliant with protocol and regulatory requirements. It achieves this through delegatable anonymous credentials (DACs) with randomization and structure-preserving signatures, and preserves privacy against DBs via unlinkable, untraceable credential presentations with updateable commitments. Moreover,

SLAPX operates with a single geolocation DB and reduces communication overhead compared to approaches requiring multiple non-colluding DBs.

- **Adaptive proximity-backed location assurance and spoofing resistance**: SLAPX introduces an adaptive PoL mechanism with two deployment modes. In infrastructure-rich settings, nearby WiFi access points (APs) or cellular towers issue time-bound PoL objects using proximity checks and event-oriented revocable-if-linked linkable ring signatures (RLRSs), enabling privacy-preserving proof acquisition and event-scoped rate limiting. In infrastructure-limited settings, users can anonymously prove proximity to a registered nearby device (ND) through a public-key distance-bounding protocol (DBP) and obtain delegated credentials that carry certified, time-sensitive PoL data for one-time use. This dual design improves robustness and resists spoofing attacks, including relay-based and distance-fraud-style attacks.

- **DoS mitigation with DB-CRN architectural compliance**: SLAPX provides proactive DoS mitigation by combining Verifiable Delay Functions (VDFs) with RLRS-based event-scoped rate limiting. It imposes verifiable computational costs on requesters while integrating puzzle setup, issuance, and validation into the spectrum-query workflow. Although puzzle-related operations are outsourced to existing DBs, SLAPX reinforces their protection through RLRS-based rate limiting, thereby mitigating abuse without violating DB-CRN architectural assumptions or regulatory constraints.

- **Practical system model and comprehensive evaluation**: We present a practical system model that integrates SLAPX into a DB-CRN architecture comprising user devices, access networks, spectrum databases, and regulatory authorities, and we provide extensive analytical, numerical, and simulation-based evaluation. Our study includes primitive-level benchmarking, communication and latency analysis, and network-level experiments for representative spoofing and DoS scenarios, offering a realistic blueprint for evaluating privacy-preserving and secure spectrum-sharing mechanisms in DB-CRNs.

1.3. Outline

The remainder of the paper is organized as follows. Section 2 presents the preliminaries, DB-CRN system architecture, and building blocks used in SLAPX. Section 3 introduces the threat model, security model, and representative attack scenarios. Section 4 describes the design, main operations, and security analysis of the SLAPX framework. Section 5 reports the primitive-level and network-level performance evaluation. Section 6 reviews related work and Section 7 concludes the paper and outlines future directions.

2. Preliminaries and building blocks

This section presents the notations, network architecture, and cryptographic building blocks of our framework.

Notations: The operators $\|$, $|x|$, and $\{0,1\}^k$ denote string concatenation, the bit-length of a value x , and the set of k -bit binary strings, respectively, while $\oplus$ represents the bitwise XOR operation. The set of natural numbers is denoted by $\mathbb{N}$. For a sequence $\{x_i\}_{i=1}^\ell$, we write $(x_1, \dots, x_\ell)$, and $\overset{\$}{\leftarrow} S$ denotes uniform random sampling from a set S . Let $\mathbb{G}_1$, $\mathbb{G}_2$, and $\mathbb{G}_T$ be cyclic groups of prime order p , and let $e : \mathbb{G}_1 \times \mathbb{G}_2 \rightarrow \mathbb{G}_T$ denote a bilinear pairing satisfying bilinearity and non-degeneracy. The i th element of a vector $\mathbf{m}$ is denoted by $\mathbf{m}[i]$, and $h(\cdot)$ represents a cryptographically secure hash function. The security parameter is denoted by λ and p denotes a large prime. Finally, sk , PK , and ID denote a secret key, a public key, and a user identifier (e.g., a MAC address), respectively, while $\text{SGN.Sign}(\cdot)$ and $\text{SGN.Verify}(\cdot)$ represent standard digital signature generation and verification algorithms.

¹ The preliminary version of our approach, namely SLAP, was presented at the IEEE SVCC 2025 Conference [14].

2.1. System architecture: DB-CRN

Network Components: The DB-CRN is a spectrum-sharing architecture in which unlicensed devices query trusted geolocation databases to obtain channel availability at their current location [5]. This model was deployed in real FCC TV White Space (TVWS) systems and adopted in the Microsoft and Google database trials [15,16]. The DB-CRN architecture consists of five key entities [17]:

- **Spectrum Regulator:** It is the regulatory authority that defines technical rules for DB-CRN operation, like incumbent protection requirements, device reporting and registration procedures, and transmit parameters [1]. This entity authorizes and oversees geolocation database administrators and provides authoritative incumbent datasets and regulatory constraints. Throughout this manuscript, the FCC is used interchangeably to denote this spectrum regulatory authority for convenience.
- **Servers:** They represent external network or cloud services (e.g., CRN, cloud, edge nodes) that users access after obtaining spectrum authorization [7]. They do not participate in spectrum assignment; instead, they simply carry user traffic over the channels approved by the PSD.
- **Private Spectrum Databases (PSDs):** PSDs are FCC-certified, cloud-based geolocation databases (e.g., Google, Microsoft, Spectrum Bridge) responsible for maintaining incumbent information, propagation constraints, and channel availability maps [18]. PSDs process spectrum-availability queries submitted by users and return permissible channels and power limits, synchronizing regularly according to FCC mandates [4].
- **User Equipments (UEs):** UEs are unlicensed devices of users (e.g., smartphones, sensors, and vehicular radios) that seek opportunistic access to licensed spectrum bands [7]. Acting as SUs, they determine their location, submit channel-availability queries to PSDs, and operate strictly within the constraints specified in PSD responses to avoid interfering with PUs. Throughout this paper, the terms client and UE are used interchangeably.
- **Access Points (APs):** APs consist of WiFi access points or cellular base stations (LTE/5G macro or small cells) that provide the IP connectivity required for UEs to reach PSDs and servers over the Internet [14]. APs do not manage spectrum availability but offer backhaul connectivity and, in some advanced architectures, assist with localization or timing support.

Communication Flow and Protocol Stack: In a DB-CRN, UEs communicate with PSDs over standard Internet paths to obtain spectrum availability, while servers and APs provide network connectivity and application-layer services. Communication exchanges follow an IP-based client-server model and use secure, lightweight protocols consistent with real FCC TVWS deployments and the IETF PAWS specification [19]. The communication workflow consists of three primary interactions:

- **UE-to-AP Communication (Access Connectivity):** UEs attach to a nearby AP to obtain Internet connectivity. This step uses standard link-layer and access-network protocols, such as IEEE 802.11 or 3GPP RRC/NAS for LTE/5G [20]. APs forward UE traffic to the Internet and do not participate in the decision-making process [19].
- **UE-to-PSD Communication (Spectrum Querying):** UEs submit spectrum-availability queries directly to PSDs using HTTPS over TLS to ensure confidentiality and integrity of location and device parameters. The request and response formats follow the JSON-based geolocation DB access protocol standardized by the IETF PAWS specification (RFC-7545) [19]. A typical UE-to-PSD exchange includes (i) a registration or device-descriptor message, (ii) a location-based *AVAIL_CHANNELS* query, and (iii) a PSD response containing permissible channels, power limits, and validity intervals. PSDs process each request using incumbent datasets and FCC rules before sending their response [20].

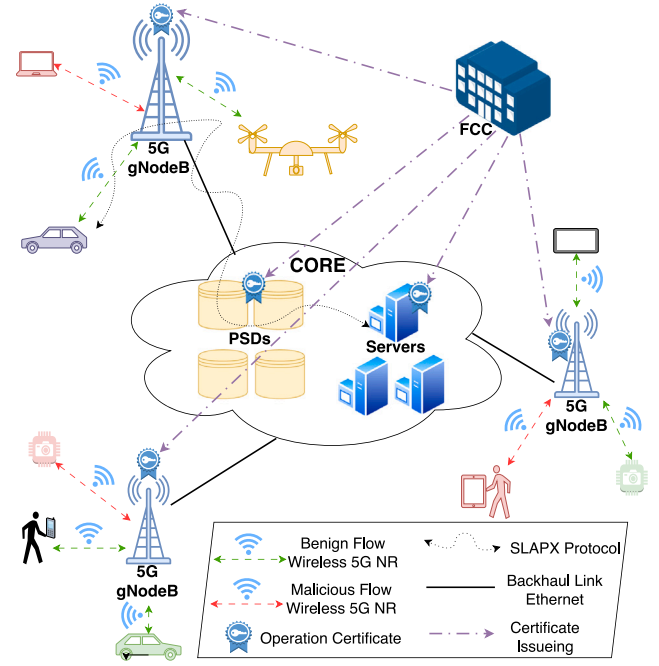


Fig. 1. Overview of DB-CRN system architecture.

- **UE-to-Server Communication (Application Traffic):** After obtaining spectrum authorization, UEs communicate with application or cloud Servers using standard IP protocols. These Servers do not participate in spectrum allocation; they merely receive traffic over the channels authorized by the PSD. Servers may verify spectrum-access credentials or tokens in enhanced architectures, but this is not required in the baseline DB-CRN model.

The communication stack follows a standard Internet protocol hierarchy, ensuring seamless interoperability with existing devices and networks. This design enables secure, fine-grained, and location-aware spectrum access without requiring any modifications to underlying access infrastructures or access points. The overall system architecture of DB-CRN is illustrated in Fig. 1.

2.2. Cryptographic building blocks

SLAPX harnesses various cryptographic components to achieve its privacy-preserving and secure operation goals:

- **Delegatable Anonymous Credentials (DAC):** We use an attribute-based DAC scheme [21] that supports multi-level attribute issuance, controlled delegation, and anonymity. The main algorithms are summarized below (see [21]).

Definition 1. A Delegatable Anonymous Credential (DAC) scheme is defined by the following algorithms:

- $(pp, sk_{RI}, PK_{RI}) \leftarrow \text{Setup}(1^\lambda, 1', 1'')$: Given the security parameter λ , a bound t on the committed attribute set size, and a delegation depth parameter $\eta > 1$, it initializes the system by producing public parameters pp and generating the root issuer's key pairs (sk_{RI}, PK_{RI}) for each level in η . pp is an implicit input to other algorithms.
- $(PK, sk) \leftarrow \text{KeyGen}(pp)$: Generates the user's key pair (PK, sk) , which PK acts as the user's initial pseudonym.
- $(nym, aux) \leftarrow \text{NymGen}(PK)$: From the user's PK , outputs a fresh pseudonym nym and auxiliary randomness aux to maintain unlinkability in subsequent interactions.

- $\text{CreateCred}(L', A, \text{sk}_{RI}) \leftrightarrow \text{GetCred}(\text{PK}_{RI}, \text{sk}_u, A) \rightarrow (\text{cred}, (\bar{C}, \bar{O}), \text{dk}_{L'})$: An interactive protocol between the root issuer (RI) and a user identified by nym_u . For the attribute set A , the RI forms a set commitment C and signs it using commitment to create a delegatable credential at PK_{RI} . The user receives the credential cred , the opening information $\bar{O}$, and a delegation key $\text{dk}_{L'}$ for level L' .
- $\text{IssueCred}(\text{PK}_{RI}, \text{dk}_{L'}, \text{sk}_u, \text{cred}_u, A_I, L'') \leftrightarrow \text{ReceiveCred}(\text{PK}_{RI}, \text{sk}_r, A_I) \rightarrow (\text{cred}_r, \text{dk}_{L''})$: It enables controlled delegation from a delegator nym_i to a recipient nym_r . Using their secret key sk_i , public key of the RI PK_{RI} , credential cred_i , delegation key $\text{dk}_{L'}$, and attribute extension A_I , the delegator generates a new credential cred_r for the recipient. The delegatee, using the secret key sk_r , obtains the credential cred_r with the updated attribute set $A' = (A, A_I)$ and delegation depth $L'' \leq L'$, along with a new delegation key $\text{dk}_{L''}$ that supports further delegation if allowed.
- $\text{CredProve}(\text{PK}_{RI}, \text{sk}_p, \text{nym}_p, \text{aux}_p, \text{cred}_p, D, \text{ctx}) \leftrightarrow \text{CredVerify}(\text{PK}_{RI}, \text{nym}_p, D, \text{ctx}) \rightarrow \{0, 1\}$: A zero-knowledge presentation protocol that allows a user to anonymously prove possession of a valid credential. The prover uses secret key sk_p , pseudonym nym_p , auxiliary randomness aux_p , and credential cred_p to generate a proof that selectively reveals only the certified disclosed attributes $D \subseteq A$. In addition, the proof may be bound to an external context ctx (e.g., beacon, timestamp, request identifier, or claimed location) so that verification is specific to the current protocol session. The verifier checks the proof under PK_{RI} with respect to both D and ctx , and outputs 1 if the proof is valid, and 0 otherwise.

• **Distance Bounding Protocol (DBP)**: A DBP is a two-party protocol that verifies the physical proximity of two entities by measuring round-trip delays in a fast challenge-response exchange. We employ a PK-DBP [22] based on a one-pass authenticated key-agreement protocol using nonce Diffie-Hellman to derive a session key between the prover and verifier. This key is then used within the OTDB symmetric-DBP [23].

Definition 2. A public key DBP is depicted the high-level algorithms: A public key DBP is defined as follows:

- $(\text{sk}, \text{PK}) \leftarrow \text{KeyGen}(1^\lambda)$: Generates a public/secret key pair for security parameter λ .
- $ss \leftarrow \text{AKA}(\text{sk}, \text{PK}, \text{PK}')$: The prover and verifier derive a shared session key using their own key pairs and the other party's public key (PK').
- $\{0, 1\} \leftarrow \text{SymDBP}(ss, th)$: An interactive proximity-test protocol using session key ss and distance threshold th , consisting of: (i) **Initialization**: The verifier samples $m \in \{0, 1\}^{2n}$ and sends it to the prover, who computes $a = ss \oplus m$. (ii) **Rapid Bit Exchange**: For $i = 1, 2, \dots, n$, the verifier sends a challenge $c_i \in \{0, 1\}$ and the prover responds with $r_i = a_{2i+c_i-1}$, while the verifier records the round-trip time RTT_i . (iii) **Authentication**: Using $a = ss \oplus m$, the measured delays, and the distance bound, the verifier checks $\text{RTT}_i \leq 2 \times th$ and $r_i = a_{2i+c_i-1}$. It outputs 1 if the prover is within the threshold; otherwise 0.

• **Revocable-iff-linked Linkable Ring Signature (RLRS)**: Ring signatures allow a user to anonymously sign on behalf of a group. An RLRS augments this by embedding an event identifier into each signature, enabling detection of multiple signatures produced by the same signer within that context and supporting controlled anonymity revocation when necessary [24].

Definition 3. A Revocable-iff-linked Linkable Ring Signature (RLRS) scheme is a 6-tuple algorithm as shown below:

- $(\text{msk}, pp_{\text{RLRS}}) \leftarrow \text{RLRS.Setup}(1^\lambda)$: Given λ , generate the master secret key msk and the public params pp_{RLRS} .

- $(\text{sk}_u) \leftarrow \text{RLRS.Extract}(pp, \text{ID}_u)$: An interactive key-extraction protocol that, given pp_{RLRS} and a user identifier $\text{ID}_u \in \{0, 1\}^*$, outputs the user secret key sk_u .
- $(\sigma, \tau) \leftarrow \text{RLRS.Sign}(\text{sk}_u, m, \mathcal{L}_{\text{ID}}, e_{\text{ID}})$: To sign a message m under identifier e_{ID} with ring $\mathcal{L}_{\text{ID}} = \{\text{ID}_1, \dots, \text{ID}_n\}$ ($n \leq t_{\text{max}}$), the signer uses a non-interactive ZK proof to produce a signature σ and link tag τ .
- $\{0, 1\} \leftarrow \text{RLRS.Verify}(\mathcal{L}_{\text{ID}}, m, e_{\text{ID}}, \sigma, \tau)$: Given the ring $\mathcal{L}_{\text{ID}}$, message m , identifier e_{ID} , and signature-tag pair (σ, τ) , output 1 if the signature is valid, or else 0.
- $\{0, 1\} \leftarrow \text{RLRS.Link}(\mathcal{L}_{\text{ID}}, e_{\text{ID}}, m, \sigma, \tau, m', \sigma', \tau')$: Determine if two signature-tag pairs (σ, τ) and (σ', τ') on messages m and m' (under event e_{ID} and ring $\mathcal{L}_{\text{ID}}$) were produced by the same signer. Return 1 if linked, or else 0.
- $(\perp, \text{ID}_u) \leftarrow \text{RLRS.Revoke}(\sigma, \sigma', e_{\text{ID}}, \mathcal{L}_{\text{ID}}, \mathcal{L}'_{\text{ID}})$: Given two linked signatures from events e_{ID} and rings $\mathcal{L}_{\text{ID}}$ and $\mathcal{L}'_{\text{ID}}$, output the signer's identity $\text{ID}_u \in \mathcal{L}_{\text{ID}} \cap \mathcal{L}'_{\text{ID}}$ if the signatures originate from the same user; output $\perp$ otherwise.

• **Verifiable Delay Function (VDF)**: A VDF [25] enforces a prescribed delay by requiring sequential computation, while enabling a fast verification of the result. We use an RSA-based VDF [26], which provides compact proofs and constant-time verification.

Definition 4. A Verifiable Delay Function (VDF) scheme is a 3-tuple algorithm as shown below:

- $(N, H, H_p) \leftarrow \text{VDF.Setup}(1^\lambda, \kappa)$: Given λ and difficulty level κ , generate an RSA modulus N of size λ , a hash function $H : \{0, 1\}^* \rightarrow \{0, 1\}^{2\kappa}$, and a prime-deriving hash $H_p(m) = \text{next-prime}(H(m))$, which returns the closest prime number greater than or equal to $H(m)$.
- $(\ell, \pi) \leftarrow \text{VDF.Eval}(m, \tau)$: On input $m \in \{0, 1\}^*$ and delay parameter $\tau \in \mathbb{N}$, compute $x = H(m)$ and the sequentially run $y = x^{2^\tau} \bmod N$. Then compute the proof using $\ell = H_p(x + y)$ and $\pi = x^{\lfloor 2^\tau / \ell \rfloor}$.
- $\{0, 1\} \leftarrow \text{VDF.Verify}(x, \tau, \ell, \pi)$: Given x, τ, ℓ, π , compute $x = H(m)$, $r = 2^\tau \bmod \ell$, and $y = \pi^r \times x^r \bmod N$. The verifier outputs 1 if $\ell = H_p(x + y)$, or else 0.

3. Threat and security models and attack scenarios

This section outlines (i) the threat model and scope of our solution, defining the adversary, its capabilities, and the considered attack vectors; (ii) the security objectives and security model that underpin the formal security proof of SLAPX as a primitive; and (iii) representative attack scenarios that motivate and guide our network-level simulations when our proposed cryptographic techniques are integrated into the network architecture.

3.1. Threat model and scope

We consider a probabilistic polynomial-time (PPT) adversary $\mathcal{A}$ that controls the public wireless communication channel and can observe, inject, modify, delay, replay, and interleave protocol messages among UEs, APs, NDs, PSDs, and service servers. The adversary may interact adaptively with honest entities, create malicious users, and orchestrate concurrent executions of the protocol. APs, PSDs, and service servers are assumed to be honest-but-curious: they follow the protocol faithfully, yet may attempt to infer sensitive information such as UE identity, credential usage patterns, or precise location from the observed transcripts. This adversarial setting is consistent with the privacy and misuse threats already targeted in the current draft, namely user deanonymization, location spoofing, and DoS.

We distinguish between *cryptographic adversarial capabilities* and *physical-layer adversarial capabilities*. On the cryptographic side, $\mathcal{A}$ may attempt to forge or misuse credentials, replay previously valid proofs, submit malformed or adaptive requests, forge puzzle tokens,

or flood PSDs and servers with resource-intensive requests. On the physical side, $\mathcal{A}$ may attempt coordinate spoofing, relay or mafia fraud, distance fraud, distance hijacking, or collusion with compromised nearby devices or access points. Accordingly, the guarantees of SLAPX are twofold: (i) cryptographic authenticity, anonymity, unlinkability, replay resistance, and rate limiting derive from the anonymous credentials, linkable ring signatures, signatures, and verifiable delay function mechanisms; and (ii) location assurance additionally depends on the fidelity of the employed proximity-testing mechanism, namely $\text{ProxVerify}(\text{RSS}, \text{RTT}, \text{env}_{\text{param}})$ in the AP-assisted mode and $\text{AKA} + \text{SymDBP}$ in the ND-assisted mode. Under this model, we consider the following attack classes:

- **Client Privacy and Anonymity Attacks:** A malicious PSD, service server, external observer, or colluding set of such parties attempts to infer the real identity of a UE, link multiple protocol sessions to the same UE, or recover sensitive certified attributes or fine-grained location information from observed credential presentations, PoL objects, or service requests. These efforts can include passive monitoring, active interaction, or correlating metadata to uncover sensitive user-specific information.
- **Location Spoofing and Proof Misuse Attacks:** A malicious user attempts to obtain spectrum access or services for a false location or outside its allowed area by submitting forged coordinates, replaying previously accepted PoL tokens, reusing delegated credentials outside their intended context, or mounting relay, mafia-fraud, distance-fraud, or distance-hijacking attacks [12]. In the AP-assisted mode, the adversary may also try to exploit the gap between cryptographic validation and physical-layer proximity estimation; in the ND-assisted mode, it may target the AKA establishment and distance-bounding subroutines.
- **Denial-of-Service Attacks:** Malicious and compromised users or external attackers may attempt to disrupt DB-CRN operations by overwhelming PSDs or service servers during spectrum queries or service requests. Such attacks can involve flooding the system with invalid, replayed, or resource-intensive requests to exhaust computational capacity, increase response latency, or deny service to legitimate users.

Scope of Our Solution: The SLAPX framework is designed to preserve user anonymity and location privacy during spectrum access. It protects SUs while they query spectrum availability, obtain cryptographic puzzles, and access network services, mitigating threats such as identity exposure, location spoofing, and DoS attacks. Our work concentrates exclusively on the spectrum query and access phases. Privacy risks arising during user registration or from spectrum usage patterns (e.g., utilization inference) fall outside the scope of this study. Although the framework is tailored for SUs, it can be extended to support PUs interacting with PSD servers; however, PU-related protections are not considered here. Moreover, SLAPX does not address location leakage during ongoing spectrum use, user mobility or handover processes [27], nor does it cover timing-based attacks, side-channel leakage, or signal-based localization methods such as triangulation [28].

3.2. Security model

Building on the threat model, we formalize the security goals of SLAPX via the following primitive- and framework-level definitions. Throughout, λ denotes the security parameter, and all adversaries are PPT unless stated otherwise.

Definition 5 (Credential Unforgeability). A DAC scheme is *unforgeable* if no adversary can cause an honest verifier to accept a presentation for a disclosed attribute set that is not derivable from any credential legitimately obtained through the issuance or delegation interfaces.

Formally, let $\text{Exp}_{\text{DAC}, \mathcal{A}}^{\text{uf}}(\lambda)$ be the experiment in which $\mathcal{A}$ is given oracle access to CreateCred , IssueCred , and presentation-related interfaces, and outputs a tuple $(\text{nym}^*, D^*, \text{ctx}^*, \pi^*)$. The adversary wins if $\text{CredVerify}(\text{PK}_{\text{RI}}, \text{nym}^*, D^*, \text{ctx}^*, \pi^*) = 1$ and there exists no honestly issued or validly delegated credential whose certified attribute set contains D^* . We require $\text{Adv}_{\text{DAC}, \mathcal{A}}^{\text{uf}}(\lambda) := \Pr[\text{Exp}_{\text{DAC}, \mathcal{A}}^{\text{uf}}(\lambda) = 1] \leq \text{negl}(\lambda)$.

Definition 6 (Credential Anonymity). A DAC scheme provides *anonymity* if no adversary can distinguish which of two honest credential holders generated an accepted presentation, even after observing adaptive protocol transcripts consistent with the security model. Formally, in experiment $\text{Exp}_{\text{DAC}, \mathcal{A}}^{\text{anon}}(\lambda)$, $\mathcal{A}$ chooses two honest users indexed by (i_0, i_1) and receives a challenge presentation generated by one of them under a hidden bit $b \leftarrow \{0, 1\}$. The adversary may continue to access all admissible oracles except those that trivially reveal the challenge signer. Its advantage is $\text{Adv}_{\text{DAC}, \mathcal{A}}^{\text{anon}}(\lambda) := \left| \Pr[b' = b] - \frac{1}{2} \right|$, and DAC is anonymous if $\text{Adv}_{\text{DAC}, \mathcal{A}}^{\text{anon}}(\lambda) \leq \text{negl}(\lambda)$.

Definition 7 (Credential Unlinkability). A DAC scheme provides *unlinkability* if no adversary can determine whether two accepted presentations were generated by the same credential holder or by two different honest holders. Formally, in experiment $\text{Exp}_{\text{DAC}, \mathcal{A}}^{\text{link}}(\lambda)$, the challenger returns two valid presentations generated either by the same user or by two different users according to a hidden bit b . The adversary outputs a guess b' , and its advantage is $\text{Adv}_{\text{DAC}, \mathcal{A}}^{\text{link}}(\lambda) := \left| \Pr[b' = b] - \frac{1}{2} \right|$. We require $\text{Adv}_{\text{DAC}, \mathcal{A}}^{\text{link}}(\lambda) \leq \text{negl}(\lambda)$.

Definition 8 (AP-Assisted PoL Correctness and Authenticity). The AP-assisted PoL mechanism is *correct* if any honest UE possessing a valid credential, an honestly generated context ctx_{TS} , and satisfying the proximity predicate of an honest AP can obtain a PoL object $\Phi_c = (m, \sigma_{\text{AP}}, \tau_{\text{AP}})$ that is accepted by the verifier except with negligible probability. It is *authentic* if no adversary can produce a fresh accepted tuple (ctx^*, Φ^*) for an event and context not endorsed by an honest AP, except with negligible probability. Formally, if $\text{Exp}_{\text{SLAPX}, \mathcal{A}}^{\text{ap-forge}}(\lambda)$ denotes the corresponding forgery experiment, then $\text{Adv}_{\text{SLAPX}, \mathcal{A}}^{\text{ap-forge}}(\lambda) := \Pr[\text{Exp}_{\text{SLAPX}, \mathcal{A}}^{\text{ap-forge}}(\lambda) = 1] \leq \text{negl}(\lambda)$. This notion captures cryptographic authenticity and event binding of the AP-generated PoL; the physical soundness of the claimed location is separately bounded by the accuracy of ProxVerify under the assumed channel and attacker model.

Definition 9 (AP-Assisted Replay and Event-Reuse Resistance). The AP-assisted PoL mechanism is *event-reuse resistant* if no adversary can cause two distinct accepted requests within the same event context to bypass the framework's replay or rate-limiting checks except with negligible probability. Formally, in experiment $\text{Exp}_{\text{SLAPX}, \mathcal{A}}^{\text{reuse}}(\lambda)$, the adversary wins if it produces two accepted requests carrying PoL objects that correspond to the same event identifier e_{ID} yet evade the framework's RLRS-based unlinkability or replay detection. We require $\text{Adv}_{\text{SLAPX}, \mathcal{A}}^{\text{reuse}}(\lambda) \leq \text{negl}(\lambda)$.

Definition 10 (ND-Assisted Delegated PoL Soundness). The ND-assisted PoL mechanism is *sound* if no adversary can obtain an accepted delegated credential cred'_c containing a spatiotemporal extension A_c unless it successfully completes the authenticated key-establishment and distance-bounding procedure with an honest nearby device and the delegated credential is validly derived from an honestly issued parent credential. Formally, let $\text{Exp}_{\text{SLAPX}, \mathcal{A}}^{\text{nd-forge}}(\lambda)$ denote the experiment in which $\mathcal{A}$ outputs a delegated credential and associated attributes accepted by the verifier without a corresponding valid delegation transcript. Then $\text{Adv}_{\text{SLAPX}, \mathcal{A}}^{\text{nd-forge}}(\lambda) := \Pr[\text{Exp}_{\text{SLAPX}, \mathcal{A}}^{\text{nd-forge}}(\lambda) = 1] \leq \text{negl}(\lambda)$.

Definition 11 (Puzzle Authenticity). The puzzle-issuance mechanism is *authentic* if no adversary can produce a fresh puzzle token and accompanying signature that are accepted as if issued by an honest PSD or server, except with negligible probability. Formally, if $\text{Exp}_{\text{SLAPX}, \mathcal{A}}^{\text{puz-auth}}(\lambda)$ denotes the puzzle-forgery experiment, then $\text{Adv}_{\text{SLAPX}, \mathcal{A}}^{\text{puz-auth}}(\lambda) \leq \text{negl}(\lambda)$.

Definition 12 (Sequential Cost Enforcement). The puzzle mechanism enforces *sequential cost* if no adversary can produce a valid puzzle solution for an honestly issued challenge while spending asymptotically less than the prescribed sequential work, except with negligible probability. Formally, in experiment $\text{Exp}_{\text{SLAPX}, \mathcal{A}}^{\text{seq}}(\lambda)$, the adversary wins if it outputs a valid solution (ℓ, π) for an honestly generated VDF challenge while incurring cost strictly below the required evaluation cost by a non-negligible margin. We require $\text{Adv}_{\text{SLAPX}, \mathcal{A}}^{\text{seq}}(\lambda) \leq \text{negl}(\lambda)$.

Definition 13 (Framework-Level DoS Resilience). The SLAPX framework is *DoS-resilient* if any adversary capable of causing an honest request to be delayed beyond a system threshold Δ must either (i) expend a proportional computational cost through the puzzle mechanism, or (ii) break at least one of the authentication, replay-resistance, or puzzle-authenticity sub-properties with non-negligible probability. Equivalently, letting $\text{Exp}_{\text{SLAPX}, \mathcal{A}}^{\text{dos}}(\lambda)$ denote the experiment in which $\mathcal{A}$ attempts to violate availability without incurring the mandated cost, we require $\text{Adv}_{\text{SLAPX}, \mathcal{A}}^{\text{dos}}(\lambda) \leq \text{negl}(\lambda)$.

3.3. Attack scenarios

In addition to the primitive-level security analysis, we consider attack scenarios for DoS and location spoofing at the network-level. These scenarios motivate and guide the network-level security simulations of the SLAPX.

3.3.1. DoS attack scenarios

We consider three representative DoS attack scenarios to capture protocol-compliant and protocol-abusive behaviors, as well as high-cost and low-cost adversarial strategies under realistic threat conditions. All scenarios are summarized in Fig. 2 and are analyzed in Section 5.2.

(1) Full Protocol DoS: The malicious users repeatedly execute the SLAPX algorithms with the goal of exhausting server-side computational and networking resources. While an honest user typically executes the protocol only once to obtain service access, the adversary continuously restarts the protocol to keep the servers busy.

(2) Computation-Bypassing DoS: The malicious users flood the server with service requests without executing the prerequisite steps of SLAPX. As a result, these requests lack valid cryptographic proofs and are discarded upon verification. However, by bypassing expensive client-side computation, adversaries can generate a significantly higher request rate, enabling large-scale flooding attacks that stress server-side filtering and rate-limiting mechanisms.

(3) Proof Precomputation DoS: The malicious users execute SLAPX up to the location proof phase and precompute a batch of valid proofs within a limited time window. Since each proof embeds a timestamp and remains valid only for a bounded duration, the attacker must carefully balance the number of precomputed proofs to maximize server load while minimizing proof expiration. This scenario captures a more strategic adversary that exploits temporal validity to amplify attack impact.

3.3.2. Location spoofing attack scenarios

We consider two representative location spoofing scenarios derived from our location verification algorithms: (i) Distance Hijacking and (ii) Distance Fraud. They capture relay-based and protocol-level proximity violations and are evaluated in Section 5.2.

(1) Distance Hijacking: A malicious UE that is physically far from the gNodeB attempts to impersonate a nearby device by relaying messages through a compromised benign UE within legitimate proximity. The benign UE is assumed to be honest but compromised, and thus serves as a relay for all protocol messages exchanged between the malicious UE and the gNodeB. By leveraging the strong received signal strength (RSS) of the compromised UE, the attacker aims to deceive the gNodeB

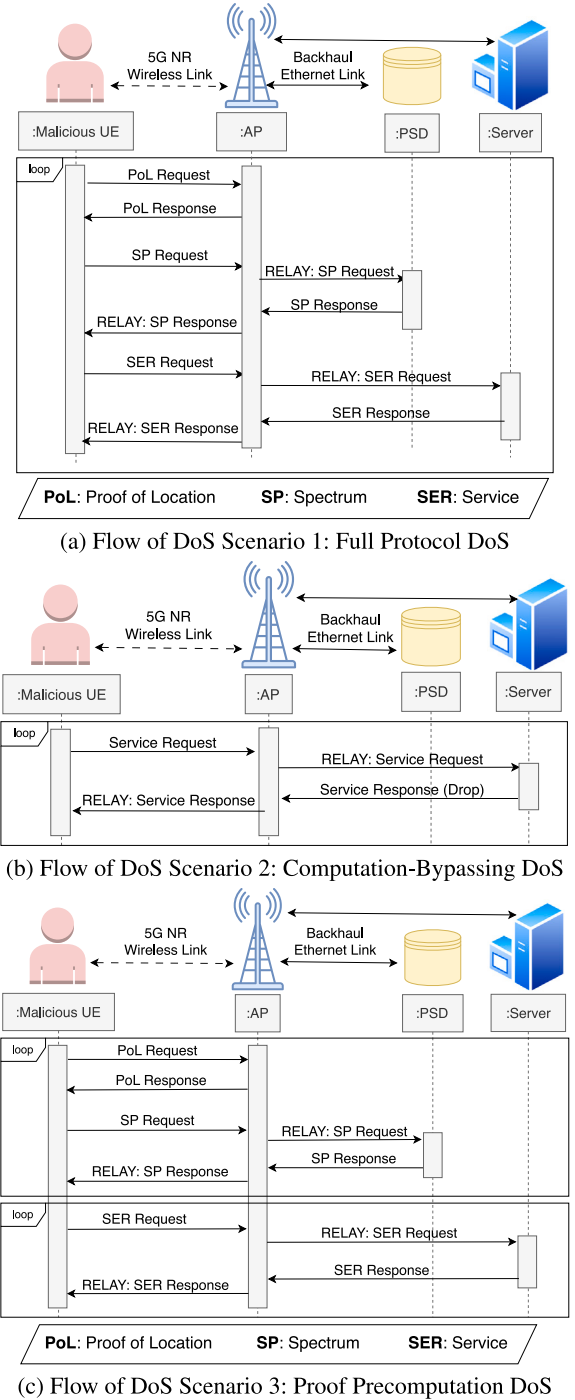


Fig. 2. SLAPX DoS attack scenarios.

into inferring a shorter distance than the malicious UE's true location. However, because the relayed communication introduces additional propagation delay, the measured round-trip time (RTT) reflects the true distance between the malicious UE and the gNodeB. This scenario allows us to study the relative effectiveness of RSS- versus RTT-based distance estimation and assess the system's resistance to relay-based proximity attacks. The message flow of this attack is shown in Fig. 3.

(2) Distance Fraud: No access point or trusted infrastructure is available in the vicinity, and proximity verification relies on device-to-device DBPs with nearby nodes. A malicious UE that is physically distant attempts to authenticate itself as nearby by manipulating the

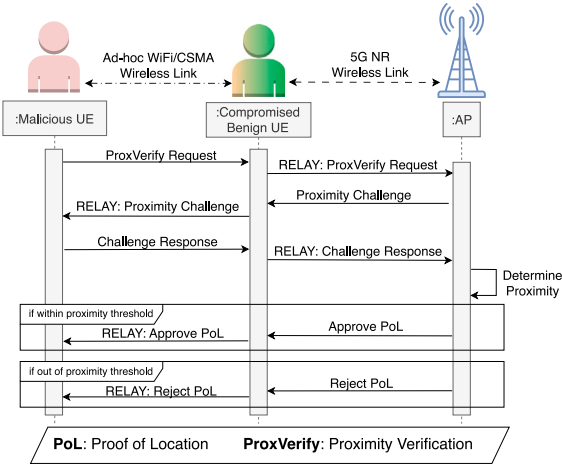


Fig. 3. Distance hijacking flow using benign UE as a relay.

DBP execution. Since the DBP enforces proximity using both RTT constraints and challenge-response correctness, the adversary cannot reduce propagation delay below physical limits. Instead, the malicious UE attempts to guess the verifier's challenges in advance and transmit corresponding responses prematurely, with the goal of appearing closer than it actually is. This scenario captures a protocol-level spoofing attack that does not rely on relaying, but instead exploits the probabilistic nature of challenge-response mechanisms under strict timing constraints.

4. SLAPX: Design and instantiation

This section presents the design and concrete instantiation of SLAPX. We first summarize the main extensions over the original SLAP framework, then describe the system setup and the operational workflow of the protocol. For readability, we present the protocol here at a high level and focus on the interactions among the UE, AP/ND, PSD, and server. The procedures are provided in Appendix, while Fig. 4 illustrates the overall algorithmic details.

Improvements over SLAP Framework and the SOTA: SLAPX extends our original SLAP framework [14] with substantial new material and the following key enhancements and additional features: (i) *Comprehensive DoS countermeasures:* We significantly strengthen DoS resilience by introducing on-demand, device-specific, and non-parallelizable VDFs, authenticated using lightweight standard signatures (e.g., ECDSA). This defense is further reinforced by a rate-limiting mechanism based on the linkability of event-oriented RLRSs, enabling effective DoS mitigation for both geolocation databases and service servers. (ii) *Enhanced Proof of Location:* We enhance the location proof mechanism by leveraging RLRS-based event linkability to enforce rate limiting on proof issuance and enable revocation of APs that collude with malicious users. This provides accountability and supports more realistic adversarial network settings. (iii) *Comprehensive performance evaluation and network simulations:* We substantially expand the evaluation by including detailed cryptographic benchmarking, comparisons with SOTA, and extensive network simulations that model real-world DoS and location spoofing scenarios under practical deployment conditions.

SLAPX advances the state of the art in DB-CRN security by providing the first unified, regulation-compliant framework that simultaneously achieves anonymous and location-private spectrum access, verifiable location assurance, and proactive DoS resilience under realistic adversarial conditions. Unlike prior solutions that address these challenges in isolation or rely on heavy cryptography, multiple non-colluding databases, or trusted infrastructure, SLAPX integrates delegatable anonymous credentials, adaptive location proofs, VDF-enforced

rate control, and RLRS-based accountability into a single lightweight architecture compatible with existing DB-CRN deployments. Compared to existing schemes, SLAPX operates with a single geolocation database, incurs orders-of-magnitude lower communication overhead, and achieves substantially lower end-to-end latency, while uniquely preserving anonymity even against access points and databases.

4.1. SLAPX framework initial setup

Database Structure: The DBs are organized in compliance with FCC spectrum-sharing regulations and maintain synchronized views of spectrum availability as required by the FCC. Each DB entry is indexed by location coordinates at a predetermined spatial resolution (e.g., 50 m), along with device characteristics (e.g., transmit power, device type) and access constraints (e.g., bandwidth/duration). This structured indexing ensures consistency across synchronized DBs. Each record includes spectrum availability data, a validity time window, and operational parameters such as the maximum allowable transmission power (EIRP).

Access Point Setup: Access Points (APs) within a geographical region jointly form a ring ($\mathcal{L}_{ID}$) that contains the ID of all participating APs. Each AP is provisioned with a cryptographic key pair (sk_{AP}, PK_{AP}), generated by the FCC using the $RLRS.Extract(pp, ID_{AP})$ algorithm (Definition 3). APs periodically broadcast time-synchronized beacons β_{TS} within predefined time windows TS to enable nearby device discovery. To verify UE proximity, each AP performs signal-based ranging using physical-layer measurements, including received signal strength (RSS) and round-trip time (RTT) [29]. These measurements are processed via $\Delta \leftarrow ProxVerify(RSS, RTT, env_{params})$, which incorporates environment-specific factors such as path-loss models, multipath effects, and channel conditions to estimate the UE-AP distance and associated confidence bounds [30]. The resulting estimate Δ determines whether the UE lies within the AP's trusted coverage region, providing a confidence interval that underpins subsequent authenticated access decisions. Additional details are provided in [31], with simulation results reported in Section 5.

System Setup: In the system initialization, the FCC, acting as the core network authority, serves as the root issuer for all participants. For each registered UE, the FCC certifies a set of device attributes and access policies by issuing a level-1 root credential via $CreateCred(L', A, sk_{FCC})$. A client, represented by a pseudonym nym_c , retrieves its credential through $GetCred(PK_{FCC}, sk_c, A)$, obtaining $(cred, (\bar{C}, \bar{O}), dk_{L'})$. The resulting credential binds the certified attribute commitment $\bar{C}$ to the FCC's public key PK_{FCC} , includes the corresponding opening information $\bar{O}$, and embeds a delegation key $dk_{L'}$ that permits controlled delegation up to level L' . In our setting, D_c denotes the selectively disclosed subset of certified attributes, including registration status, UE class, maximum transmission power, and credential expiration time (i.e., $D_c = (reg_c, Client_{Class}, MTP, t_{exp})$), while ctx denotes external session-specific context bound to the presentation proof. Such context may include fresh values such as beacons, timestamps, request identifiers, or claimed coordinates. Unless explicitly delegated as part of an attribute extension A_e , these context values are not embedded into the credential itself.

4.2. SLAPX framework main operations

Figs. 5 and 4 depict the high-level benign flow of the SLAPX protocol and the detailed algorithmic procedure, respectively. At a high level, the UE first obtains a proof of location, then uses it to anonymously query the PSD for spectrum availability, and finally evaluates the issued VDF before submitting spectrum-usage notifications or service requests to the servers. The detailed algorithmic realization of these procedures is provided in Appendix; here, we describe the protocol flow and the role of each building block at the system level.

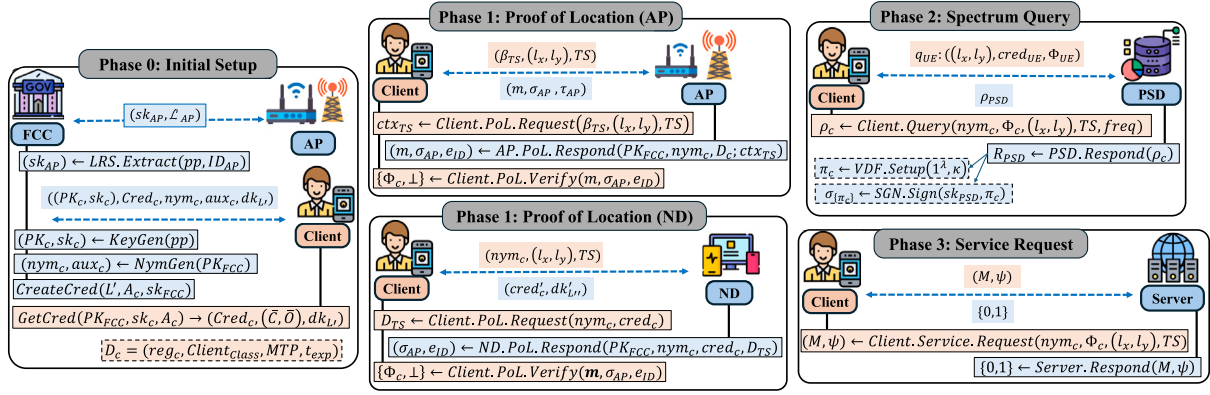


Fig. 4. SLAPX algorithmic procedure.

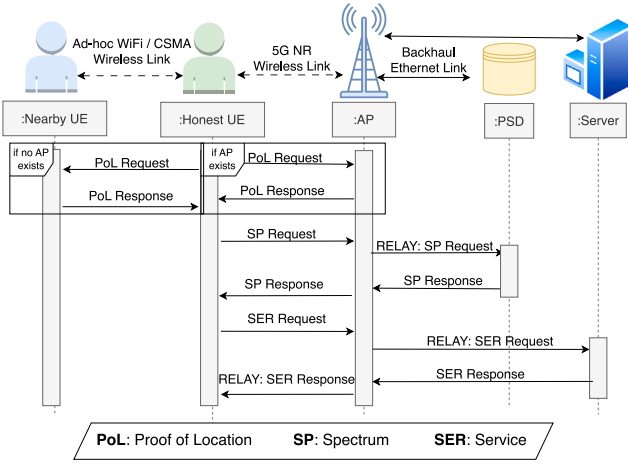


Fig. 5. High-level flow of the SLAPX protocol.

4.2.1. Proof of location acquisition and validation

Before querying PSDs or accessing network services, a UE must first obtain a valid proof of location (PoL). In SLAPX, the PoL is bound to a specific geographic claim and time window, and is later used to support anonymous yet accountable spectrum access. The framework supports two complementary PoL generation modes: an AP-assisted mode for infrastructure-rich environments and a device-assisted mode for resource-constrained settings where no nearby AP is available.

(i) *AP-Assisted PoL Generation*: When a nearby AP is available, the client first receives a beacon β_{TS} and forms a session-specific context $\text{ctx}_{TS} = (\beta_{TS}, (l_x, l_y), TS)$. It then anonymously proves possession of a valid FCC-issued credential while selectively disclosing only the certified attributes required for the current interaction, e.g., $D_c = (\text{reg}_c, \text{Client}_{\text{class}}, \text{MTP}, t_{\text{exp}})$, with the proof bound to ctx_{TS} . Thus, the beacon, claimed coordinates, and time window are treated as fresh session context rather than as static credential attributes. After validating the UE's presentation under the FCC's public key, the AP assesses the UE's proximity using physical-layer measurements through $\text{ProxVerify}(\cdot)$ to obtain the proximity scalar (th). If the claimed location lies within the resulting confidence region (Δ_{th}), the AP constructs a message tuple containing the UE's disclosed attributes, pseudonym, and session context, derives an event identifier e_{ID} , and produces an RLRS signature over this tuple. The resulting object $\Phi_c = (m, \sigma_{AP}, \tau_{AP})$ serves as the AP-assisted proof of location. The UE verifies this RLRS signature locally before accepting Φ_c for subsequent query and access operations.

(ii) *Device-Assisted PoL Generation*: In environments with limited infrastructure, such as rural areas without nearby WiFi APs or cellular

coverage, the UE may instead obtain a PoL with the assistance of a neighboring device (ND). As in the AP-assisted case, the UE forms a session context $\text{ctx}_{TS} = ((l_x, l_y), TS)$ and anonymously proves possession of its FCC-issued credential while selectively disclosing the certified attributes required for validation. After verifying the UE's presentation, the ND establishes a shared session key through an authenticated key-agreement procedure and executes the symmetric DBP to test whether the UE lies within the allowed proximity threshold. If this check succeeds, the ND creates an attribute extension $A_l = ((l_x, l_y), TS)$ and issues a delegated credential cred'_c to the UE with restricted delegation rights. The UE then retrieves this delegated credential and forms an updated attribute set A'_c that embeds the newly certified spatiotemporal information through A_l . Hence, unlike the AP-assisted case, where location and time remain part of externally bound proof context, the device-assisted mode incorporates them directly into the delegated credential.

4.2.2. Querying spectrum availability and services

Each spectrum query in SLAPX is preceded by a valid PoL corresponding to the UE's claimed position and the current time window. Once a valid PoL has been obtained, the UE anonymously proves possession of an appropriate credential and submits a spectrum-availability query to the PSD for the target location and channel request.

In the *AP-assisted* case, the UE uses its FCC-issued credential together with the AP-generated Φ_c . The credential presentation selectively reveals the required certified UE attributes and is bound to a query-specific context that includes the PoL and query parameters. The PSD first validates the UE's anonymous credential presentation, then verifies the RLRS signature embedded in Φ_c . To detect replay and misuse within the same event window, the PSD checks event consistency and evaluates RLRS linkability against other proofs observed in the current interval. If the proof is valid and unlinkable to any previously accepted conflicting request, the PSD generates a VDF instance with an appropriate difficulty parameter, signs the corresponding challenge information, and returns the spectrum response together with the VDF and its signature.

In the *device-assisted* case, the UE instead relies on the delegated credential cred'_c , whose updated attribute set A'_c already embeds the relevant spatiotemporal information. The UE proves possession of this delegated credential and submits the same type of spectrum query to the PSD. After validating the delegated credential, the PSD follows the same response logic by issuing a VDF challenge, signing it, and returning the resulting spectrum availability information to the UE. In both cases, the PSD learns only the information necessary for policy-compliant spectrum authorization, while the UE's identity remains hidden behind its anonymous credential presentation.

4.2.3. Notifying spectrum usage and service requests

After receiving the PSD's response, the UE evaluates the issued VDF through sequential computation and derives a token bound to the message M , where M encodes either a spectrum-usage notification or a network-service request. The UE then sends (M, ψ) together with a fresh anonymous credential presentation to the PSD/server. If the UE is operating in the *AP-assisted* mode, it proves possession of its FCC-issued credential while binding the presentation to the service message M , and the verifier additionally checks the associated AP-generated Φ_c . If the UE is operating in the *device-assisted* mode, it proves possession of the delegated credential cred'_c with updated attributes A'_c , which already encode the corresponding spatiotemporal information.

Upon receiving the request, the PSD or server first verifies the authenticity of the VDF challenge information and then validates the VDF solution submitted by the UE. It next checks the corresponding credential presentation and, depending on the mode of operation, validates either the RLRS-backed AP-assisted PoL or the delegated credential obtained from the device-assisted procedure. If all checks succeed, the PSD records the reported spectrum usage or the server grants the requested service. Distinct from prior approaches, SLAPX protects not only the initial spectrum query but also subsequent usage reporting and service access through anonymous credentials and verifiable access tokens. This preserves UE privacy while still enabling policy enforcement, accountability, and resistance to replay and DoS-oriented abuse.

4.3. Security analysis

Based on the established threat and security models, we now state the main security theorem for SLAPX. The proof proceeds through a sequence of hybrid games and simulation arguments, reducing any successful attack on SLAPX to either breaking one of the underlying hardness assumptions of the instantiated primitives or violating the binding and uniqueness properties of the framework's context- and event-derivation mechanisms. The supporting lemmas, full experiments, and detailed reductions are deferred to [Appendix A.2](#).

Security Assumptions: Let λ be the security parameter, and let $\mathcal{A}$ be any PPT adversary interacting with SLAPX. Assume that: (i) the DAC instantiation used by SLAPX satisfies its issuance-, delegation-, and presentation-security requirements under the hardness assumptions underlying SPSEQ-UC [32] and its commitment randomization and public-key switching mechanisms, namely the generic-group security of the construction together with the DDH-style indistinguishability assumptions used to prove origin-hiding, derivation-privacy, and conversion-privacy; (ii) the RLRS instantiation is secure in the random oracle model under the Discrete Logarithm (DL), Decisional Diffie-Hellman in the target setting used by the construction (DDL), and q -Strong Diffie-Hellman (q -SDH) assumptions; (iii) the authenticated key-establishment and public-key distance-bounding components used in the ND-assisted mode satisfy their stated security guarantees under the corresponding Diffie-Hellman-type assumptions (e.g., GDH for the attached instantiation), and the rapid challenge-response phase achieves the stated fraud-resistance bound; (iv) the signature scheme used to authenticate puzzle issuance is EUF-CMA secure under its underlying hardness assumption; (v) the VDF enforces sequential evaluation under the repeated-squaring assumption in the chosen group of unknown order, and its proof system is sound under the corresponding unknown-order-group assumption used by the instantiation; (vi) the hash functions used for event derivation and context binding are collision resistant and second-preimage resistant.

Theorem 1. *Under the security assumptions, the SLAPX framework satisfies the following security guarantees except with negligible probability in λ : (i) Anonymous and unlinkable UE authentication: no adversary can produce an accepted credential presentation for unauthorized attributes, distinguish which honest UE generated a challenge presentation, or link two*

accepted presentations beyond negligible advantage. More precisely, there exist PPT adversaries B_1, B_2, B_3 such that $\text{Adv}_{\text{SLAPX}, \mathcal{A}}^{\text{auth}}(\lambda) \leq \text{Adv}_{B_1}^{\text{DAC-uf}}(\lambda) + \text{Adv}_{B_2}^{\text{DAC-anon}}(\lambda) + \text{Adv}_{B_3}^{\text{DAC-link}}(\lambda) + \text{negl}(\lambda)$. (ii) AP-assisted PoL authenticity and event-scoped replay resistance: no adversary can produce a fresh accepted AP-assisted PoL without honest AP endorsement, nor reuse PoL objects within the same event context while evading the framework's RLRS-based replay and linkability checks. More precisely, there exist PPT adversaries B_4, B_5, B_6, B_7 such that $\text{Adv}_{\text{SLAPX}, \mathcal{A}}^{\text{ap-pol}}(\lambda) \leq \text{Adv}_{B_4}^{\text{RLRS-uf}}(\lambda) + \text{Adv}_{B_5}^{\text{RLRS-link}}(\lambda) + \text{Adv}_{B_6}^{\text{Hash-coll}}(\lambda) + \text{Adv}_{B_7}^{\text{Hash-2pre}}(\lambda) + \text{negl}(\lambda)$. (iii) ND-assisted delegated PoL soundness: no adversary can obtain an accepted delegated spatiotemporal credential extension without either breaking the authenticated key-establishment/distance-bounding subprotocols or forging a valid delegated credential chain. More precisely, there exist PPT adversaries B_8, B_9, B_{10}, B_{11} such that $\text{Adv}_{\text{SLAPX}, \mathcal{A}}^{\text{nd-pol}}(\lambda) \leq \text{Adv}_{B_8}^{\text{AKA}}(\lambda) + (3/4)^n + \text{Adv}_{B_9}^{\text{DAC-uf}}(\lambda) + \text{Adv}_{B_{10}}^{\text{Hash-coll}}(\lambda) + \text{Adv}_{B_{11}}^{\text{Hash-2pre}}(\lambda) + \text{negl}(\lambda)$. (iv) Authenticated and cost-enforcing DoS resilience: no adversary can cause accepted query or service requests to bypass the framework's per-request computational cost and authentication checks without either forging authenticated puzzle material, violating VDF sequentiality or soundness, or breaking event-scoped replay/rate-limiting enforcement. Precisely, there exist PPT adversaries $B_{12}, B_{13}, B_{14}, B_{15}, B_{16}, B_{17}$ such that $\text{Adv}_{\text{SLAPX}, \mathcal{A}}^{\text{dos}}(\lambda) \leq \text{Adv}_{B_{12}}^{\text{SIG-EUF}}(\lambda) + \text{Adv}_{B_{13}}^{\text{VDF-seq}}(\lambda) + \text{Adv}_{B_{14}}^{\text{VDF-snd}}(\lambda) + \text{Adv}_{B_{15}}^{\text{RLRS-link}}(\lambda) + \text{Adv}_{B_{16}}^{\text{Hash-coll}}(\lambda) + \text{Adv}_{B_{17}}^{\text{Hash-2pre}}(\lambda) + \text{negl}(\lambda)$.

Additionally, Section 5.2 presents a network-level security analysis of SLAPX, including simulation-based evaluations of its resilience to various DoS and location spoofing attack scenarios in wireless network settings.

5. Performance evaluation

This section provides a comprehensive evaluation of the SLAPX framework, covering both cryptographic primitive-level and network-level performance analysis.

5.1. Primitive-level evaluation

We evaluate the primitive-level performance of SLAPX by first describing the evaluation metrics and implementation setup, followed by an empirical analysis of the framework and a comparison with state-of-the-art schemes.

5.1.1. Configuration and experimental setup

Hardware: We evaluate the efficiency of SLAPX on a standard desktop platform equipped with an Intel Core i9 – 11900K @3.50 GHz, 64 GiB RAM, 1 TB SSD running Ubuntu 22.04.4 LTS.

Libraries: Our evaluation combines custom implementations and standard public cryptographic libraries across multiple languages, including C, Python, and Java. In particular, we use *DAC-from-EQS*² for DAC, *Wesolowski-VDF*³ for VDFs, *RELIC*⁴ for pairing- and elliptic-curve-based operations, and *OpenSSL*⁵ for standard cryptographic primitives such as hash functions, modular arithmetic, and exponentiation. To support comparison baselines, we additionally use *percy++*⁶ for multi-server PIR components, *liboqs*⁷ for post-quantum primitives, and *NTL*⁸ for lattice-based puzzle components. Database instances were constructed

² <https://github.com/mir-omid/DAC-from-EQS>.

³ <https://github.com/futxor/Wesolowski-VDF>.

⁴ <https://github.com/relic-toolkit/relic>.

⁵ <https://www.openssl.org/>.

⁶ <https://percy.sourceforge.net/>.

⁷ <https://openquantumsafe.org/>.

⁸ <https://libntl.org/>.

Table 1
Computational and communication overhead of SLAPX.

Phase	Entity	Computational Cost (ms)	Communication Overhead (B)
PoL.AP	Client	32.2	2456
	AP	75.79	
PoL.ND	Client	31.8	1944
	ND	77.05	
Spectrum Query	Client	17.23	3016
	PSD	74.34	
Service Request	Client	$\kappa \times S_q + 17.23$	2712
	Server	77.68	

Note: S_q and κ represent the repeated squaring time to solve a puzzle and puzzle difficulty, respectively.

using *SQLite*⁹ to ensure a consistent and reproducible backend across all evaluated settings.

Configuration and Parameter Selection: We exclude signal transmission and network communication delays from our analysis, as these are typically constant-time operations and occur at the microsecond scale. Our configuration employs SHA-256 for hashing, structure-preserving set commitments [32], SPSEQ-UC signatures [21], and the BN-256 elliptic curve (EC) for binding and EC operations (with $n = 2048$). Anonymous authentication is realized via Schnorr-type zero-knowledge proofs augmented with Damgård’s technique, following the DAC construction [21]. Non-interactive proofs are instantiated via the Fiat–Shamir transform, yielding an effective security level of approximately 100 bits. The SGN digital signature is instantiated using ECDSA, a widely adopted and computationally efficient standard signature scheme.

Evaluation Metrics and Rationale: We empirically analyze the SLAPX framework by measuring cryptographic, computational, and communication overhead across all operational phases and underlying primitives, including DAC, RLRS, DBP, and VDF. Since no existing solution provides an equivalent combination of functionalities, direct performance comparisons are not applicable. Instead, we present a comprehensive performance breakdown of SLAPX along key metrics to assess its practicality and deployability. We further include a qualitative and analytical comparison with representative schemes that address partial aspects of these functionalities in the DB-CRN setting. The evaluation proceeds as outlined below.

5.1.2. Experimental results

The empirical evaluation of cryptographic overhead, computational costs, and communication overhead for each phase of the SLAPX framework is summarized in Table 1 and detailed below:

Cryptographic Overhead: To demonstrate credential ownership, the user re-randomizes both the credential cred and pseudonym nym , and executes a ZKPoK over the secret key sk and auxiliary randomness aux . This process yields a fresh, unlinkable pseudonym and selectively reveals an attribute subset D via a set commitment construction. In practice, the dominant runtime overhead arises from signature conversion, representation normalization, and re-binding to a new set commitment, which require approximately 2 ms, 5 ms, and 13 ms, respectively. In the adopted PK-DBP, the interactive AKA phase incurs one ECC multiplication (0.61 ms), a single hash evaluation (0.35 ms), and random value generation (0.05 ms). The subsequent rapid bit-exchange phase operates at nanosecond granularity; the resulting distance-fraud error margin (on the order of 1–10 m) is negligible relative to the remaining protocol costs. Proximity verification at the AP, implemented via RSS and RTT measurements in $\text{ProxVerify}(\cdot)$, typically completes within 1–10 ms depending on client distance and environmental conditions.

⁹ The database is modeled as a matrix with adjustable row counts (e.g., 2^{10} , 2^{12} , 2^{14} , and 2^{16}).

When instantiated using ECDSA, the SGN operation requires about 0.3 ms for signature generation and 3.8 ms for verification. For the RLRS component, signing, verification, linking, and revocation require approximately 15.8 ms, 14.98 ms, 0.12 ms, and 0.6 ms, respectively. These costs can be further reduced at the PSD through batch verification and parallelized linking. Finally, solving a VDF on commodity hardware incurs increasing latency with higher difficulty parameters κ : about 17 ms for 10^3 squarings, 61 ms for 5×10^3 , 121 ms for 10^4 , 874 ms for 8×10^4 , and 3.17 s for 3×10^5 . In contrast, VDF setup and verification remain constant-time, averaging roughly 89 ms and 4 ms, respectively, across all difficulty levels.

Computational Costs: (i) *PoL.AP Phase:* The client proves its credential cred_c and verifies the RLRS signature, while the AP validates the credential, executes ProxVerify , and issues an RLRS signature as the location proof. (ii) *PoL.ND Phase:* Two users engage in interactive protocols that include credential proving and verification, key agreement AKA, symmetric-DBP, and credential delegation with the location proof embedded as a certified attribute. (iii) *Query Phase:* The client submits a valid PoL (from either an AP or an ND) to query the PSD for spectrum availability at a given location and time. The PSD verifies the location proof and linkability (via RLRS or cred_c), then returns spectrum data together with a user-specific VDF. (iv) *Service.Request Phase:* The client evaluates the VDF, proves possession of a valid credential, and submits the result. The server (or PSD) verifies the solution before authorizing service access or updating the DB.

Communication Overhead: We consider groups of size $|\mathbb{G}_1| = |\mathbb{Z}_p| = 256$ bits, $|\mathbb{G}_2| = 512$ bits, and $|\mathbb{G}_T| = 3072$ bits, with arithmetic performed modulo a 2048-bit integer. Message payloads satisfy $|m| < 256$ B, timestamps occupy 8 B on a 64-bit Unix system, precise location coordinates require 16 B, and spectrum availability data β , derived from FCC records, is approximately 560 B. All attributes in the system are assumed to have uniform size. Credentials encapsulate the values $|\text{cred}|$, $|\text{sk}|$, and $|\text{nym}|$ within set commitments and SPSEQ-UC signatures, yielding a constant-size representation that does not depend on the number of attributes. Specifically, the credential size is $4|\mathbb{G}_1| + |\mathbb{G}_2| + |\mathbb{Z}_p|$, totaling 1792 bits. The size of the commitment vector $\vec{C}$ scales with the delegation depth (here $L = 2$), resulting in communication overhead that grows linearly with both the number of attributes and delegation levels. The RLRS signature is independent of the number of APs and contains $\sigma = 2|\mathbb{G}_1| + 3\mathbb{G}_1 + 15\mathbb{Z}_p = 640$ bytes. Using publicly available FCC database records,¹⁰ we estimate each spectrum database entry to be roughly 560 bytes, augmented with synthetic data for evaluation. The aggregate communication costs across all protocol phases are summarized in Table 1.

Qualitative and Analytical Comparison with SOTA: Table 2 compares SLAPX against representative DB-CRN schemes across both functionality and efficiency dimensions. The compared works span peer-to-peer anonymity mechanisms, pseudonym-based designs, PIR-based spectrum-query schemes, blockchain- or EPID-assisted solutions, and recent privacy-preserving DoS-aware frameworks. As shown in the table, most prior schemes address only a subset of the desired properties. In particular, pseudonym- and PIR-based constructions mainly target query privacy, but typically do not provide anonymous PoL acquisition, event-scoped misuse detection, or integrated counter-DoS protection. Similarly, schemes that incorporate some form of location verification often rely on auxiliary entities or explicit verifier-side knowledge of user context, whereas DoS-aware designs such as PACDoSQ [9] provide request-throttling support but do not integrate privacy-preserving location assurance into the same workflow. In contrast, SLAPX is the only scheme in Table 2 that jointly supports anonymous spectrum access, location privacy, proximity-backed location assurance under

¹⁰ <https://enterpriseefiling.fcc.gov/dataentry/public/tv/lmsDatabase.html>.

Table 2

Qualitative and analytical comparison with existing location privacy schemes.

Scheme	Features						Delay				Total communication
	Setting	Security assumption	Loc.Privacy	Anonym	Loc.Verification	Counter-DoS	SU	PSD	E2E	PoL	
Troja et al. [33]	1-DB	IT-PIR, DL (AV-net)	Peer-to-Peer	AV-net (GA)	X	X	1650 ms	11760 ms	13410 ms	X	12 MB
Li et al. [34]	1-DB	DL, CDH, ElGamal	Pseudo-ID	X	WiFi AP+Loc.Server	X	X	X	X	210 ms	X
Xin et al. [8]	1-DB	QRA, RSA, AES	PIR	X	WiFi AP+QRA	X	292.8 ms	142.7 ms	407.4 ms	430.1 ms	325 kB
LP-Chor [35]	c-DB	IT	PIR	X	X	X	7.7 ms	480 ms	620 ms	X	753 kB
LP-Goldberg [35]	c-DB	IT, RSC	PIR	X	X	X	320 ms	1210 ms	1780 ms	X	6 MB
RAID-LP-Chor [35]	c-DB	IT, PRG	PIR	X	X	X	0.4 ms	22 ms	210 ms	X	125 kB
Zeng et al. [36]	1-DB	BS+ECC	DL/CDH	PseudoID	X	X	87 ms	27 ms	135 ms	X	1.24 kB
TrustSAS [4]	c-DB	IT, RSA	PIR	EPID	X	X	329.4 ms	324.6 ms	4954 ms	X	1.25 MB
PACDoSQ [9]	c-DB	IT, M-LWE, SHA-256	PIR	Tor	X	HBP	28.1 ms	199 ms	1373.6 ms	X	605.92 kB
SLAPX	1-DB	DDH, q-SDH, RSA	DAC	DAC	WiFi-AP+RLRS DBP+DAC	VDF Rate-Limiting	17.23 ms	74.34 ms	78.61 ms	107.99 ms 108.85 ms	3.72 kB

Libraries: The reported comparison values were obtained under a unified evaluation setup using public implementations and standard libraries, including *percy++* for multi-server PIR, *liboqs* for post-quantum primitives, and *OpenSSL* for standard cryptographic operations. For all schemes in Table 2, the numerical values were derived by our own implementation or controlled reproduction under the same desktop environment and parameter settings whenever possible. The exception is TrustSAS, for which the blockchain-related component was derived following the methodology and reported cost model in the original paper, consistent with the way that scheme is evaluated by its authors. **Variables:** We consider six databases for multi-DB schemes with $|DB| = 560$ MB and 400 rows/columns as described in [8]. **Symbols:** Key terms include BS (base station), HBP (hash-based puzzles), QRA (quadratic residue assumption), EPID (enhanced privacy ID based on direct anonymous attestation), AV-net (Anonymous Veto Network), DL (Discrete Logarithm), CDH (Computational Diffie-Hellman), IT (Information-Theoretic security), q-SDH (q-Strong Diffie-Hellman), ToR (The Onion Routing), and M-LWE (Module Learning With Errors). X indicates not supported.

two deployment modes, and proactive DoS mitigation within a practical single-DB architecture.

(i) *End-to-End Spectrum Query Delay:* Among the compared schemes, several PIR-based designs achieve strong query privacy but incur substantial end-to-end latency due to database-side processing and, in some cases, multi-database coordination. For example, LP-Chor, LP-Goldberg, RAID-LP-Chor, TrustSAS, and PACDoSQ report end-to-end delays of approximately 620 ms, 1780 ms, 210 ms, 4954 ms, and 1373.6 ms, respectively, while Xin et al. report 407.4 ms [4,8,9,35]. By comparison, SLAPX achieves an end-to-end spectrum-query delay of only 78.61 ms with a client-side cost of 17.23 ms and PSD-side cost of 74.34 ms. This reduction follows from avoiding multi-server PIR and heavy query-obfuscation procedures while still preserving anonymous and unlinkable access.

(ii) *Location-Assurance Cost:* Only a small subset of prior schemes incorporates any explicit location-verification mechanism. Li et al. and Xin et al. report PoL costs of about 210 ms and 430.1 ms, respectively, while most other schemes omit location verification entirely [8,34]. In contrast, SLAPX provides two privacy-preserving PoL modes with total acquisition costs of 107.99 ms for the AP-assisted mode and 108.85 ms for the nearby-device-assisted mode. Thus, SLAPX not only enables stronger functionality than the compared schemes, but also achieves substantially lower PoL latency than prior location-aware designs.

(iii) *Communication Overhead:* The communication cost of prior approaches varies widely. Multi-DB PIR-based constructions incur particularly large overhead, including 753 KB for LP-Chor, 6 MB for LP-Goldberg, 125 KB for RAID-LP-Chor, 1.25 MB for TrustSAS, and 605.92 KB for PACDoSQ [4,9,35]. Even single-database schemes such as Troja et al. and Xin et al. require 12 MB and 325 KB, respectively [8,33]. In comparison, SLAPX requires only 3.72 KB total communication overhead while operating with a single geolocation database and supporting both anonymous authentication and PoL validation. This highlights a key practical advantage of SLAPX: it avoids the communication costs of PIR-based privacy while still preserving strong privacy and accountability properties.

(iv) *Architectural Simplicity and Coverage of Security Properties:* Several prior schemes rely on multiple synchronized DBs, dedicated location servers, or separate anonymity layers, which complicate deployment and widen the trust surface. By contrast, SLAPX preserves a lightweight single-DB architecture while integrating credential-based anonymous access, RLRS-backed AP-assisted PoL, DBP-based nearby-device-assisted PoL, and VDF-based request throttling into one coherent workflow. Thus, beyond improving latency and communication overhead, SLAPX offers a more complete and deployment-oriented security architecture than existing DB-CRN schemes.

5.2. Network-level evaluation

This section presents an instantiation of the SLAPX protocol in a realistic wireless network setting and evaluates its performance through

Table 3

Simulation parameters in the ns3 experiments.

Parameter	Value
Channel model	3GPP TR 38.901 (default)
UE Tx Power	30 dBm
gNodeB Tx Power	30 dBm
Number of cells per gNodeB	3
Beam width	50 degrees
Backhaul data rate	1 Gbps
Backhaul delay	2 ms
MTU (DoS)	1500 bytes
MTU (Fragmentation)	576–9000 bytes

simulations. We use ns3 [37] to model the deployment of SLAPX and analyze its communication overhead, while the 5G NR (New Radio) LENA module [38] is employed to simulate the 5G infrastructure.

5.2.1. Simulation setup and configuration

Overview of the ns3 Setup: We assume that the regulatory authority (e.g., the FCC) has already provisioned all required certificates, keys, and credentials. As a result, the FCC is not explicitly modeled in the ns3 environment. The remaining entities in the SLAPX framework are mapped as follows: the client is modeled as a 5G NR User Equipment (UE), the AP as a 5G gNodeB, and the PSD as a server node. Furthermore, servers that provide cloud services are modeled as one single server node. In this setup, each UE operates within the radio access network (RAN) and connects to the nearest gNodeB over a wireless channel using 5G NR links. Server nodes are deployed in the core network and are connected to the gNodeBs via wired backhaul links.

5G Parameter Configuration: The 5G NR LENA module offers a wide range of configuration options for realistic 5G simulations. For simplicity, we retain default settings for most parameters and manually configure only a selected subset. The configured parameters and their values are summarized in Table 3. Specifically, the channel model follows the standard NR specification [39] used by default in the LENA module. The transmission power of both UE and gNodeB nodes is set to 30 dBm, reflecting typical deployment conditions. The Maximum Transmission Unit (MTU) is set to the standard Ethernet frame size of 1500 bytes, and the backhaul network is modeled with a data rate of 1 Gbps and an end-to-end latency of 2 ms.

Server Modeling: Because ns3 is a discrete-event simulator, directly modeling real-time server overload caused by DoS attacks is non-trivial. While network-level congestion can be simulated, capturing the impact of DoS attacks on server-side processing requires an abstracted model. Therefore, each server is represented using a queue-and-worker abstraction, a common approach in analytical DoS modeling [40]. Specifically, the server consists of two components: (i) *Workers*, which model independent computational units (e.g., CPU cores) capable of

Table 4

Symbols used in the scope of DoS experiments.

Symbol	Description
N_U	Total number of UEs spawned
N_Q	Total number of messages that are queued
N_D	Total number of dropped benign messages due to busy workers and full queue capacity
t_Q	The average waiting time of a message in the queue (milliseconds)
r_{mal}	The ratio of malicious UEs to the total number of UEs

processing requests in parallel; and (ii) *Queue*, which buffers incoming requests when all workers are busy. However, when the queue is full, incoming requests are dropped. This model enables us to capture server saturation and request loss under DoS conditions, serving as the basis for evaluating DoS scenarios in our simulations.

Terminology: To present the evaluation results clearly and concisely, we use a set of symbols to represent the performance metrics and experiment parameters used in our DoS scenarios, as shown in Table 4.

5.2.2. Simulation results

Fragmentation Analysis of SLAPX: Based on a realistic 5G deployment, we evaluate SLAPX through numerical analysis focusing on fragmentation behavior and communication overhead. Due to the relatively large size of certain protocol messages, packet fragmentation becomes observable as illustrated in Fig. 6, where the upper illustration shows the relationship between MTU values and fragmentation across different SLAPX message types. Although the 5G standard does not mandate a fixed MTU, many deployments adopt the standard Ethernet MTU of 1500 bytes [41]. Under this setting, only the *Service Request* and *Spectrum Request* messages experience fragmentation, increasing from one to two packets per message. The lower plot in Fig. 6 further analyzes communication overhead under varying MTU sizes, illustrating the proportion of protocol and network headers relative to payload size. As expected, messages with smaller payloads incur higher relative overhead due to header dominance.

DoS Experiments: Each of the three attack cases, along with a baseline scenario without SLAPX, is evaluated independently. For each scenario, we simulate 10 s of network time, with the attack initiated at $t = 2$ and terminated at $t = 8$.

Baseline DoS: To establish a baseline and expose system vulnerabilities under adversarial load, we first evaluate the network in the absence of any protection mechanism. In this baseline scenario, SLAPX is disabled and malicious UEs directly flood the server with requests to induce service disruption. We conduct a large set of experiments with varying N_U and r_{mal} . System performance is evaluated using three key metrics: (i) the number of queued messages (N_Q), (ii) the average delay experienced by messages in the queue (t_Q), and (iii) the number of requests from benign UEs that are dropped due to full queue capacity and busy workers (N_D).

Among the evaluated performance metrics, Fig. 7(a) illustrates the number of requests that cannot be processed immediately and are therefore placed into the server queue, denoted by N_Q . Apparently, N_Q increases sharply with both N_U and r_{mal} , reaching nearly 2000 messages in environments with high ratio of malicious UEs. Note that this value reflects the cumulative number of messages enqueued over time, rather than the maximum queue length at any instant. This distinction arises because queued messages are continuously dequeued for processing as workers become available while new requests arrive concurrently. Similar trends are observed in Fig. 7(b), which reports the average waiting time of messages in the server queue, representing the time a request spends queued before being processed by an available worker and denoted by t_Q . Without anti-DoS countermeasures provided by SLAPX, t_Q exceeds 100 ms in most scenarios and can rise to nearly

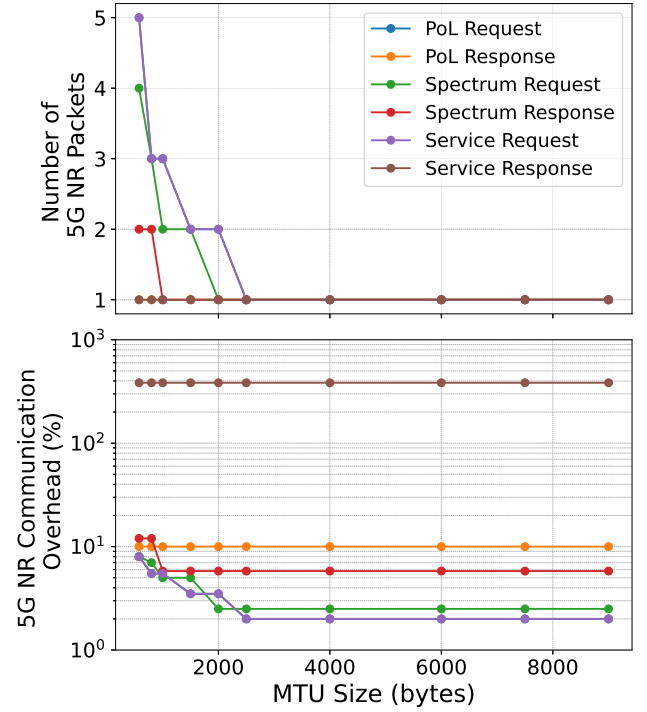


Fig. 6. Fragmentation analysis for SLAPX protocol messages with varying MTU sizes.

500 ms as N_U and r_{mal} increase. Such delays can severely impact time-sensitive applications and degrade overall service quality.

The most severe performance degradation is observed in Fig. 7(c), which shows the number of benign requests dropped by the server due to the DoS attack, denoted by N_D . As expected, N_D increases with N_U , since a larger population generates more legitimate traffic. Notably, the highest drop rate occurs when r_{mal} is 20%, exceeding even the $r_{mal} = 40\%$ case. This behavior can be explained by the reduction in the number of benign UEs as the malicious fraction grows, which ultimately limits the volume of legitimate traffic subject to dropping. This finding displays the vulnerability of a regular SAS without any counter-DoS mechanism.

Full Protocol DoS: We conduct experiments with varying N_D and r_{mal} , and summarize the results in Fig. 8. System performance is evaluated using three metrics: N_Q , t_Q , and N_D . As shown in Fig. 8(a), SLAPX decreases N_Q to fewer than 50 messages during the simulations, a significant reduction compared to the baseline scenario, which yielded 1000 messages. Consistently, t_Q remains below 65 ms, as illustrated in Fig. 8(b). Moreover, no benign message drops are observed ($N_D = 0$), demonstrating that the DoS attack is effectively mitigated by SLAPX under this scenario.

Computation-Bypassing DoS: Again, we conduct experiments with varying N_U and r_{mal} , with results shown in Fig. 9. The system is evaluated using the same three performance metrics as in the previous scenario. As illustrated in Fig. 9(b), t_Q remains almost negligible and is even lower than in the *Full-Protocol DoS* scenario, showcasing the effectiveness of SLAPX. While Fig. 9(a) shows a relatively larger N_Q , this increase is expected and inevitable because adversaries can send many more requests without having to perform the computational challenge. Nevertheless, in this scenario, the server promptly identifies and discards malicious requests that lack valid puzzle solutions, preventing them from consuming processing resources. The effectiveness of this early filtering is further confirmed by the consistently low t_Q observed across all configurations, indicating that SLAPX does not incur significant performance degradation in this scenario either. Still, one can place a network-level firewall in front of the application and

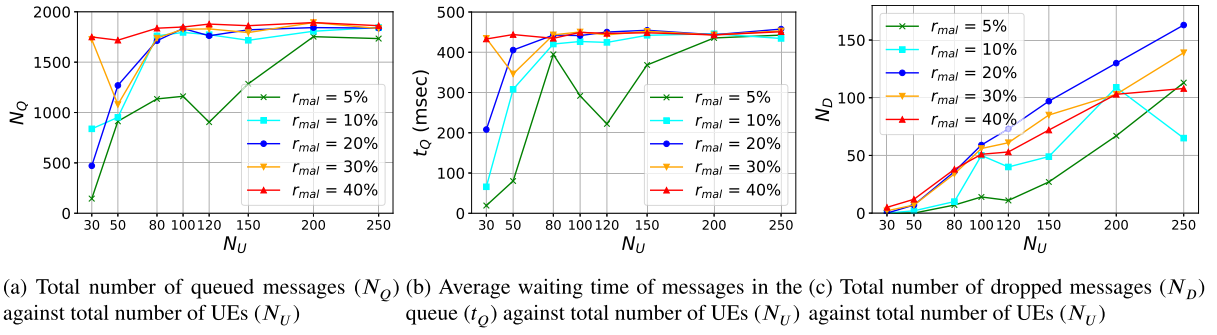


Fig. 7. Performance metrics for the Baseline DoS scenario across different numbers of UEs and malicious UE ratios.

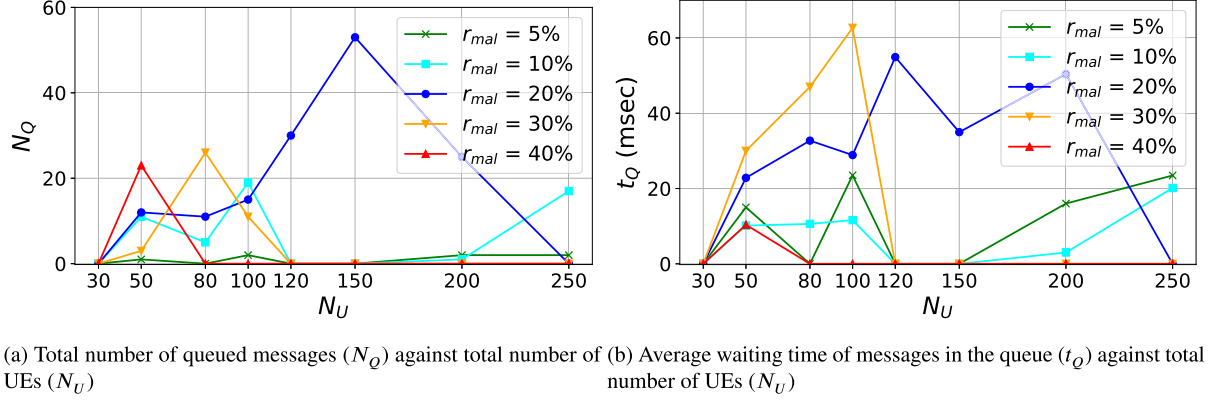


Fig. 8. Performance metrics across different numbers of UEs and malicious UE ratios, in the Full Protocol DoS scenario.

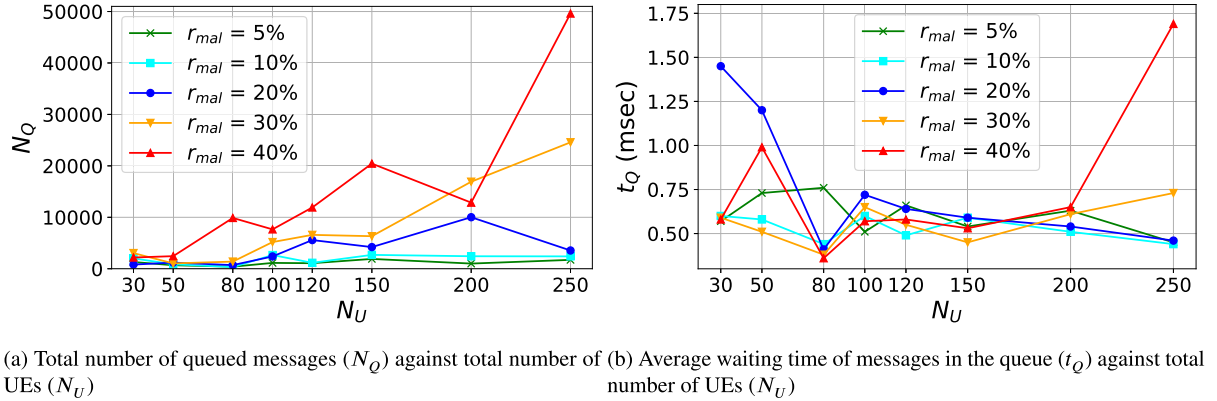


Fig. 9. Performance metrics across different numbers of UEs, in the Computation-Bypassing DoS scenario.

add customized rules to drop high-frequency messages, from the same source, on a network level. This additional mechanism would prevent the server queue from filling quickly and thus, helps to obtain a much lower N_Q .

To further explain the large number of queued messages observed in this scenario, we examine the maximum queue length reached during the simulations, as shown in Fig. 11. This analysis was omitted for the Full-Protocol DoS scenario due to the consistently low N_Q . In contrast, the present scenario exhibits a higher volume of queued requests, warranting closer inspection of queue saturation. As illustrated, the queue capacity of 100 is reached only in two configurations—both with 250 UEs and r_{mal} of 30% and 40%. Although packet drops occur in these cases, all dropped packets originate from malicious UEs, and no benign requests are denied service. This result confirms that SLAPX effectively contains the attack and preserves service availability for legitimate users.

Proof Precomputation DoS: Again, we conduct experiments with varying N_U and r_{mal} , with results displayed in Fig. 10. The system is evaluated using the same three performance metrics, with results shown in Figs. 10(a) and 10(b). As observed, system performance closely resembles that of the previous protected scenarios and remains significantly better than the baseline configuration without SLAPX. Notably, zero benign messages are dropped ($N_D = 0$), demonstrating that SLAPX successfully prevents service disruption across all three DoS attack scenarios.

In addition to the standard metrics evaluated across all DoS scenarios, the puzzle pre-computation attack introduces an additional dimension of analysis. Although malicious UEs may attempt to pre-compute multiple puzzles in advance, the effectiveness of such an attack is inherently bounded. Each proof of location (PoL) includes a timestamp and is valid only within a limited time window. As a result, puzzle solutions expire after a fixed duration, and excessive

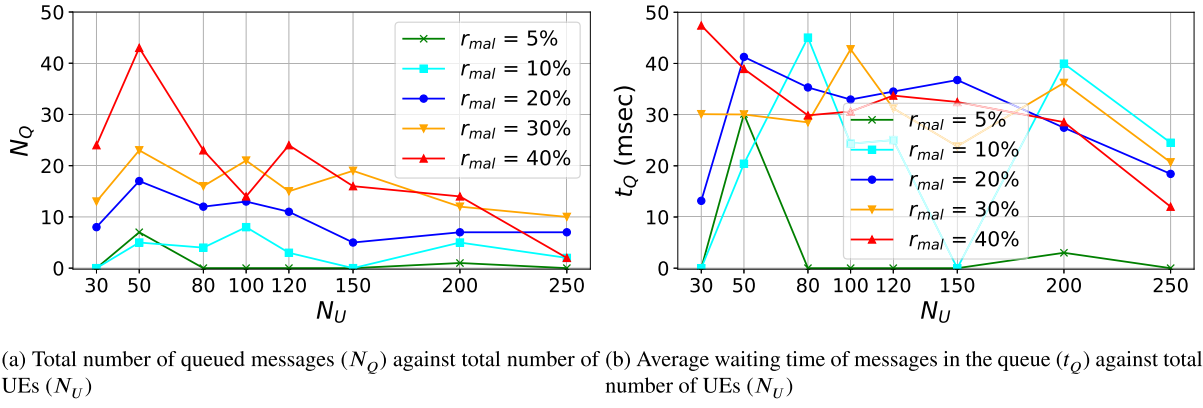


Fig. 10. Performance metrics across different numbers of UEs, in the *Proof Precomputation DoS* scenario.

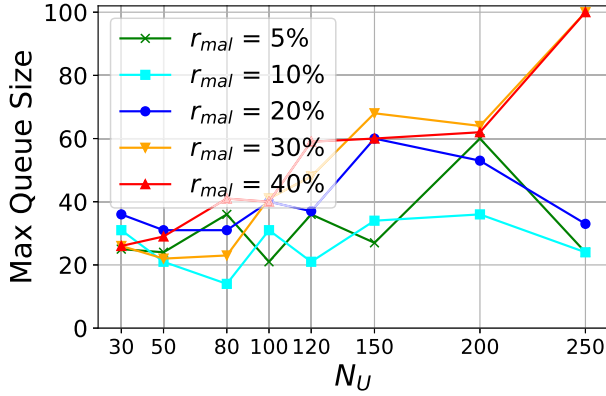


Fig. 11. Maximum queue size against total number of UEs (N_U), in the *Computation-Bypassing DoS* scenario.

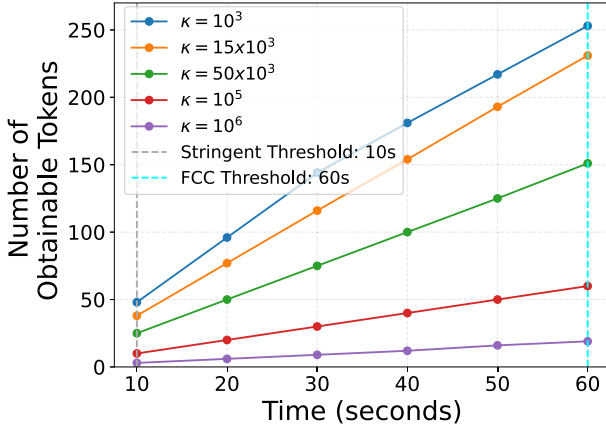


Fig. 12. Limits of the proof pre-computation process.

pre-computation causes earlier solutions to become invalid before they can be used. Consequently, the maximum number of usable tokens an adversary can accumulate is constrained by the PoL expiration threshold and the puzzle difficulty parameters. This trade-off is illustrated in Fig. 12. As illustrated, no UE can pre-compute more than 250 puzzles, even under the easiest configuration and with the FCC-defined validity threshold of 60 s. If the system operator adopts a stricter expiration window or increases puzzle difficulty, the number of obtainable tokens drops sharply. This analysis confirms that the defense provided by SLAPX can withstand adversaries with strong computational capabilities.

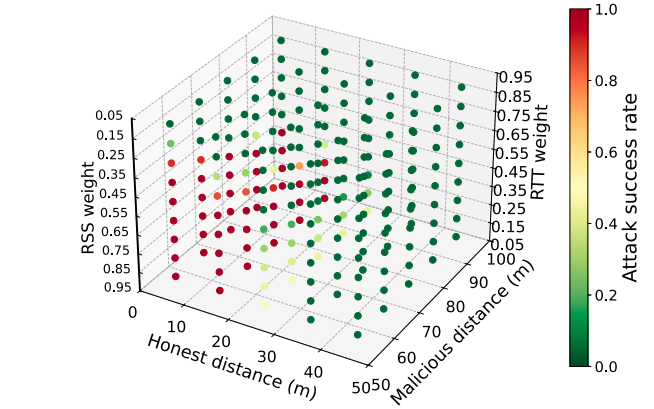


Fig. 13. 3D Scatter chart for attack success rate in the distance hijacking attack.

Table 5
Parameters of distance hijacking simulations.

Parameter	Value range	Value step
Honest UE distance	0–50 m	10 m
Malicious UE distance	50–100 m	10 m
RTT weight in ProxVerify	0.1–0.9	0.1

Location Spoofing Experiments: Here, we present the results of our experiments on the two location spoofing scenarios, each of which is evaluated separately.

Distance Hijacking: For one of the location spoofing scenarios, we evaluate the effectiveness of the Proximity Verification function (*ProxVerify*), which enables a gNodeB to estimate the distance to UEs using a combination of received signal strength (RSS) and round-trip time (RTT) measurements. In the considered distance hijacking attack, a malicious UE communicates with a compromised benign UE over an ad-hoc WiFi network using CSMA, while the benign UE maintains a legitimate 5G NR connection with the gNodeB. The benign UE is configured to relay all messages between the malicious UE and the gNodeB without delay. The proximity threshold is set to 50 m, in accordance with FCC guidelines [42], such that devices within this range are classified as proximate, while those beyond it are considered distant.

To investigate the success of relay attacks under multiple factors, we vary a set of configuration parameters listed in Table 5. For each configuration, the experiment is repeated 100 times and the corresponding attack success rate is recorded. Fig. 13 presents a 3D scatter plot in which color encodes the attack success rate as a function of the honest UE distance to the gNodeB, the malicious UE distance, and the relative

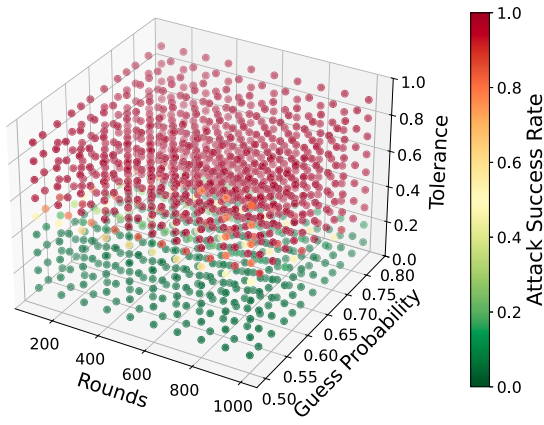


Fig. 14. 3D Scatter chart for attack success rate in the distance fraud attack. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

weighting of RTT versus RSS in the *ProxVerify* function. The results show that attack success increases as the honest UE moves closer to the gNodeB and as the influence of RSS outweighs that of RTT. This observation aligns with earlier results, indicating that stronger emphasis on RTT improves resistance against distance hijacking. Moreover, as the honest UE approaches the gNodeB, the resulting higher RSS can be exploited by the malicious UE, further increasing the likelihood of a successful relay attack.

Distance Fraud: In this scenario, we examine the DBP, which enables a client to authenticate using a nearby, already authenticated device when no access point is available. We consider one honest UE that is legitimately connected to the network and one malicious UE attempting to mislead the nearby device into accepting it as being closer than its actual distance. The DBP enforces proximity by executing multiple rounds of rapid challenge–response exchanges. In each round, the device must return a sequence of bits in the correct order and within a strict time bound determined by the measured RTT. In this setup, we initiate the distance fraud scenario by first placing an authenticated, honest UE in the network and then introducing a malicious UE at a random distance between 50 and 100 m from it. Since it is not possible to exceed the speed of light, the malicious device cannot reduce RTT to appear closer. However, the adversary can guess the challenge bits in advance and respond before receiving them. If the guesses are correct, the malicious UE may succeed in authenticating despite being outside the allowed proximity range.

To model the DBP and characterize the attacker’s capabilities, we consider the following parameters:

- **Rounds:** The total number of challenge bits transmitted by the verifier (i.e., the honest UE) to the malicious UE during the DBP execution.
- **Tolerance:** The fraction of rounds in which a device is permitted to respond incorrectly while still being authenticated. For example, with a tolerance of 0.2 and 100 rounds, authentication succeeds if at least 80 rounds are answered correctly.
- **Guess Probability:** The probability that a malicious UE correctly predicts a challenge bit before receiving it. While this probability is 50% under random guessing, we conservatively allow higher values to model an attacker with partial predictive advantage.

To evaluate the success probability of distance fraud attacks, Fig. 14 presents a 3D scatter plot in which color denotes the attack success rate as a function of the number of rounds, the attacker’s guess probability, and the DBP tolerance threshold. As expected, attack success increases with higher tolerance levels and greater guess probabilities, since both parameters favor the adversary. Nevertheless, when the tolerance is

set to a reasonable value—such as 10% or lower—the probability of a successful distance fraud attack becomes negligible.

6. Related work

We review prior work on privacy-preserving spectrum access, location verification, and DoS mitigation in DB-CRNs from a scheme-level perspective.

(i) Privacy-Preserving Spectrum Access: Existing approaches can be broadly categorized into anonymization-based, perturbation-based, and cryptographic techniques. k -anonymity and pseudonym-based schemes [43] rely on identity hiding within anonymity sets but provide limited guarantees in dynamic or sparse deployments. Differential privacy and location obfuscation techniques [44] introduce controlled noise into user queries to protect sensitive information, at the cost of reduced accuracy in spectrum availability decisions. Private Information Retrieval (PIR)-based approaches [4,8,35] enable query privacy by preventing databases from learning user interests, but incur significant computational and communication overhead and often assume multi-server settings. More recent works explore credential-based designs that use anonymous credentials and selective disclosure to enable unlinkable authentication and access control, offering stronger privacy guarantees with improved practicality.

(ii) Location Verification and Proof-of-Location: Location assurance mechanisms in DB-CRNs typically rely on infrastructure-based, hardware-assisted, or cooperative protocols. Infrastructure-based solutions leverage trusted entities such as APs or location servers to validate user proximity, often requiring dense deployment or strong trust assumptions [8]. Hardware-assisted approaches leverage specialized components (e.g., secure modules or GPS-based attestation) to ensure location integrity, but introduce deployment and cost constraints. Cooperative and distance-bounding protocols enable proximity verification through peer interactions or signal timing analysis, providing resilience against relay and distance-fraud attacks but often requiring synchronized communication and additional protocol complexity. Most existing designs expose fine-grained location information to verifiers or lack support for unlinkable and privacy-preserving proofs.

(iii) DoS Mitigation in DB-CRN Systems: DoS defenses in spectrum access systems span detection-based, access-control-based, and cryptographic rate-limiting approaches. Intrusion detection systems (IDSs), including ML-based techniques [2], identify abnormal traffic patterns but require continuous monitoring and are susceptible to adversarial evasion. Blockchain-based access control mechanisms provide decentralized accountability but introduce latency and scalability challenges. Client Puzzle Protocols (CPPs) [45,46] impose computational effort on requesters prior to service access, mitigating request flooding; however, large-scale deployments face bottlenecks in puzzle generation, distribution, and parallel execution. Outsourced puzzle generation reduces server overhead but shifts the attack surface to other system components, such as DBs.

7. Conclusion and future work

DB-CRNs improve spectrum utilization but introduce fundamental security and privacy challenges stemming from mandatory location disclosure, vulnerability to location spoofing, and exposure to DoS attacks on centralized infrastructure. In this work, we presented SLAPX, a unified and regulation-compliant security framework that simultaneously addresses these challenges under realistic adversarial assumptions. SLAPX enables privacy-preserving and anonymous spectrum access, provides adaptive and verifiable location proofs without revealing precise user location, and ensures robust DoS resilience through verifiable delay functions combined with RLRS-based rate limiting. Comprehensive cryptographic evaluation and network simulations demonstrate that SLAPX achieves significantly lower latency and communication

overhead than existing approaches while effectively resisting spoofing and DoS attacks. As SLAPX currently targets classical security, future work will focus on extending its resilience to quantum-capable adversaries and providing long-term post-quantum security guarantees for DB-CRNs. Moreover, we plan to strengthen the location-assurance component beyond purely cryptographic guarantees.

CRediT authorship contribution statement

Saleh Darzi: Writing – review & editing, Writing – original draft, Project administration, Methodology, Investigation, Formal analysis. **Gökcan Cantali:** Writing – review & editing, Writing – original draft, Software, Investigation, Formal analysis. **Attila Altay Yavuz:** Writing – review & editing, Supervision, Project administration, Methodology, Investigation, Funding acquisition. **Gürkan Gür:** Writing – review & editing, Supervision, Investigation, Funding acquisition.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Attila Altay Yavuz reports financial support was provided by National Science Foundation. Gurkan Gur reports financial support was provided by National Science Foundation. Attila Altay Yavuz has patent System and method for secure locationproof & anonymous privacy-preserving spectrum access pending to University of South Florida St Petersburg. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

A.1. Cryptographic algorithms

The detailed cryptographic algorithms of the proposed SLAPX scheme are provided in Algorithms 1, 2, and 3. Algorithm 1 describes the AP-assisted PoL acquisition. The client generates a presentation proof over the selectively disclosed attribute set $D_c = (\text{reg}_c, \text{Client}_{\text{Class}}, \text{MTP}, t_{\text{exp}})$, while binding fresh session context $\text{ctx}_{\text{TS}} = (\beta_{\text{TS}}, (l_x, l_y), \text{TS})$ to the proof. Upon successful credential verification, the AP performs proximity validation and constructs a location- and time-bound message m incorporating both D_c and ctx_{TS} . It then produces an event-oriented LRS over m , where the event identifier e_{ID} binds the proof to the current location-time context. The client verifies the signature and outputs Φ_c , which serves as a verifiable, privacy-preserving PoL token for subsequent interactions.

Algorithm 2 captures the ND-assisted PoL acquisition through controlled delegation. The client proves possession of a valid credential over the disclosed attributes D_c , binding only minimal context $\text{ctx}_{\text{TS}} = ((l_x, l_y), \text{TS})$. After successful verification and proximity confirmation via authenticated key agreement and distance bounding, the ND issues a delegated credential cred'_c by embedding the location-time tuple as an attribute extension A_l . Unlike the AP-assisted case, the location proof is incorporated into the credential itself through delegation rather than external context binding. The client verifies and derives (cred'_c, A'_c) , where A'_c augments the original attribute set with the delegated location proof, enabling subsequent authentication without requiring external PoL tokens.

Algorithm 1 $\Phi_c \leftarrow \text{PoL.AP}(\beta_{\text{TS}}, (l_x, l_y), \text{TS}, \text{nym}_c)$

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1  $\text{ctx}_{\text{TS}} \leftarrow \text{Client.PoL.Request}(\beta_{\text{TS}}, (l_x, l_y), \text{TS})$ 
2: Given the latest beacon  $\beta_{\text{TS}}$ :
3: Set  $D_c \leftarrow (\text{reg}_c, \text{Client}_{\text{Class}}, \text{MTP}, t_{\text{exp}})$ 
4: Set  $\text{ctx}_{\text{TS}} \leftarrow (\beta_{\text{TS}}, (l_x, l_y), \text{TS})$ 
5:  $\text{CredProve}(\text{PK}_{\text{FCC}}, \text{sk}_c, \text{nym}_c, \text{aux}_c, \text{cred}_c, D_c; \text{ctx}_{\text{TS}})$ 
6:  $(m, \sigma_{\text{AP}}, \tau_{\text{AP}}) \leftarrow \text{AP.PoL.Respond}(\text{PK}_{\text{FCC}}, \text{nym}_c, D_c; \text{ctx}_{\text{TS}})$ 
7: if  $l = \text{AP.CredVerify}(\text{PK}_{\text{FCC}}, \text{nym}_c, D_c; \text{ctx}_{\text{TS}})$ 
8:    $\Delta_c \leftarrow \text{AP.ProxVerify}(\text{RSS}, \text{RTT}, \text{enu}_{\text{param}})$ 
9:   if  $(l_x, l_y) \in \Delta_c$ 
10:    Set  $m \leftarrow (D_c, \text{nym}_c, \text{ctx}_{\text{TS}})$ 
11:    Set  $e_{\text{ID}} \leftarrow H(\mathcal{L}_{\text{ID}}, (l_x, l_y) \| \text{TS} \| \beta_{\text{TS}})$ 
12:     $(\sigma_{\text{AP}}, \tau_{\text{AP}}) \leftarrow \text{RLRS.Sign}(\text{sk}_{\text{AP}}, m, \mathcal{L}_{\text{ID}}, e_{\text{ID}})$ 
13:    Send  $(m, \sigma_{\text{AP}}, \tau_{\text{AP}})$  to the Client.
14:  $\{\Phi_c, \perp\} \leftarrow \text{Client.PoL.Verify}(m, \sigma_{\text{AP}}, \tau_{\text{AP}})$ 
15: if  $l = \text{RLRS.Verify}(\mathcal{L}_{\text{ID}}, m, e_{\text{ID}}, \sigma_{\text{AP}}, \tau_{\text{AP}})$ 
16:   Return  $\Phi_c \leftarrow (m, \sigma_{\text{AP}}, \tau_{\text{AP}})$ 

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Algorithm 2 $(\text{cred}'_c, A'_c) \leftarrow \text{PoL.ND}(\text{TS}, (l_x, l_y), \text{nym}_c)$

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1  $\text{ctx}_{\text{TS}} \leftarrow \text{Client.PoL.Request}(\text{nym}_c, (l_x, l_y), \text{TS})$ 
2: Request PoL and a delegated credential  $\text{cred}'_c$  from an ND
3: Set  $D_c \leftarrow (\text{reg}_c, \text{Client}_{\text{Class}}, \text{MTP}, t_{\text{exp}})$ 
4: Set  $\text{ctx}_{\text{TS}} \leftarrow ((l_x, l_y), \text{TS})$ 
5:  $\text{CredProve}(\text{PK}_{\text{FCC}}, \text{sk}_c, \text{nym}_c, \text{aux}_c, \text{cred}_c, D_c; \text{ctx}_{\text{TS}})$ 
6:  $(\text{cred}'_c, \text{dk}'_{L''}, A_l) \leftarrow \text{ND.PoL.Respond}(\text{ctx}_{\text{TS}}, \text{nym}_c)$ 
7: if  $l = \text{ND.CredVerify}(\text{PK}_{\text{FCC}}, \text{nym}_c, D_c; \text{ctx}_{\text{TS}})$ 
8:   Perform  $ss \leftarrow \text{AKA}(\text{sk}_{\text{ND}}, \text{PK}_{\text{ND}}, \text{PK}_c)$ 
9:   if  $l = \text{SymDBP}(ss, th)$  and  $(l_x, l_y) \in \Delta_{th}$ 
10:    Set  $A_l \leftarrow ((l_x, l_y), \text{TS})$ 
11:    Set  $\text{dk}'_{L''} := \perp$ 
12:    IssueCred( $\text{PK}_{\text{FCC}}, \text{dk}'_{L'}$ ,  $\text{sk}_{\text{ND}}, \text{cred}_{\text{ND}}, A_l, L''$ )
13: Send  $(\text{cred}'_c, \text{dk}'_{L''})$  to the client.
14:  $(\text{cred}'_c, A') \leftarrow \text{Client.PoL.Verify}(\text{cred}'_c, A_l)$ 
15:  $(\text{cred}'_c, \text{dk}'_{L''}) \leftarrow \text{ReceiveCred}(\text{PK}_{\text{FCC}}, \text{sk}_c, A_l)$ 
16: Set  $A'_c \leftarrow (A, \Phi_c)$  where  $\Phi_c \leftarrow A_l$ 
17: Return  $(\text{cred}'_c, A'_c)$ 

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Algorithm 3 presents the full SLAPX protocol, integrating PoL acquisition, spectrum query, and service request. The client first obtains a PoL either via AP-assisted (Φ_c) or ND-assisted (cred'_c, A'_c) mechanisms, and generates a presentation proof over the disclosed attributes (D_c or A'_c), binding query-specific context ctx_c when applicable. Upon receiving the request, the PSD verifies both the credential proof and the PoL: in the AP-assisted case, it validates the ring signature and enforces event-scoped linkability to prevent replay or duplicate queries within the same context; in the ND-assisted case, it verifies the delegated credential directly. The PSD then returns a VDF puzzle along with a signed token. The client solves the puzzle to generate a PoW, which is submitted with a service request. Finally, the server verifies the puzzle solution, credential proof, and PoL validity before granting access, ensuring DoS resilience, location assurance, and privacy-preserving authentication.

A.2. Security proofs

We now prove the security guarantees of SLAPX using a sequence of hybrid games and simulation arguments. Throughout, λ denotes the security parameter, all adversaries are PPT, and $\text{negl}(\lambda)$ denotes a negligible function in λ . For each claim, we define the real experiment, gradually replace concrete protocol components by idealized or simulated ones, and show that any non-negligible gap between successive

Algorithm 3 SLAPX Scheme

1 $\rho_c \leftarrow \text{Client.Query}(\text{nym}_c, \Phi_c, (l_x, l_y), \text{TS}, \text{freq})$

1: Given (l_x, l_y) and TS, client request PoL:
2: **if** $\Phi_c \leftarrow \text{Client.PoL.AP}(\beta_{\text{TS}}, (l_x, l_y), \text{TS}, \text{nym}_c)$
3: Set $\text{ctx}_c \leftarrow (\Phi_c, \text{ch}, \text{TV})$
4: $\text{CredProve}(\text{PK}_{\text{FCC}}, \text{sk}_c, \text{nym}_c, \text{aux}_c, \text{cred}_c, D_c; \text{ctx}_c)$
5: Query a PSD for $\rho_c \leftarrow ((l_x, l_y), \text{ch}, \text{TV})$
6: **elseif** $(\text{cred}'_c, A') \leftarrow \text{Client.PoL.ND}(\text{TS}, (l_x, l_y), \text{nym}_c)$
7: $\text{CredProve}(\text{PK}_{\text{FCC}}, \text{sk}_c, \text{nym}_c, \text{aux}_c, \text{cred}'_c, A'_c)$
8: Query a PSD for $\rho_c \leftarrow ((l_x, l_y), \text{ch}, \text{TV})$

2 $R_{\text{PSD}} \leftarrow \text{PSD.Respond}(\rho_c)$

9: **if** $1 = \text{CredVerify}(\text{PK}_{\text{FCC}}, \text{nym}_c, D_c; \text{ctx}_c)$
10: Parse $(m, \sigma_{\text{AP}}, \tau_{\text{AP}}) \leftarrow \Phi_c$
11: **if** $1 = \text{RLRS.Verify}(\mathcal{L}_{\text{ID}}, m, e_{\text{ID}}, \sigma_{\text{AP}}, \tau_{\text{AP}})$
12: **for all** queries (m', σ', τ') in current TS **do**
13: $e'_{\text{ID}} \leftarrow H(\mathcal{L}_{\text{ID}}, (l_x, l_y) \parallel \text{TS} \parallel \beta_{\text{TS}})$
14: **if** $e'_{\text{ID}} = e_{\text{ID}}$
15: **if** $1 = \text{RLRS.Link}(\mathcal{L}_{\text{ID}}, e_{\text{ID}}, m, \sigma_{\text{AP}}, \tau_{\text{AP}}, m', \sigma', \tau')$
16: **return** $\perp$
17: **else**
18: Set $\pi_c \leftarrow \text{VDF.Setup}(1^\lambda, \kappa)$ accordingly
19: $\sigma_{\pi_c} \leftarrow \text{SGN.Sign}(\text{sk}_{\text{PSD}}, \pi_c)$
20: **return** $R_{\text{PSD}} \leftarrow (\pi_c, \sigma_{\pi_c}, \text{freq})$
21: **elseif** $1 = \text{CredVerify}(\text{PK}_{\text{FCC}}, \text{nym}_c, A'_c)$
22: Set $\pi_c \leftarrow \text{VDF.Setup}(1^\lambda, \kappa)$ accordingly
23: $\sigma_{\pi_c} \leftarrow \text{SGN.Sign}(\text{sk}_{\text{PSD}}, \pi_c)$
24: **return** $R_{\text{PSD}} \leftarrow (\pi_c, \sigma_{\pi_c}, \text{freq})$

3 $(M, \psi) \leftarrow \text{Client.Service.Request}(\text{nym}_c, \Phi_c, (l_x, l_y), \text{TS})$
Notifying spectrum usage or requesting services to Servers:

25: Given the request/notification message as M
26: $(\ell, \psi) \leftarrow \text{VDF.Eval}(M, \tau)$
27: **if** using AP-assisted PoL
28: $\text{CredProve}(\text{PK}_{\text{FCC}}, \text{sk}_c, \text{nym}_c, \text{aux}_c, \text{cred}_c, D_c; M)$
29: **elseif** using ND-assisted PoL
30: $\text{CredProve}(\text{PK}_{\text{FCC}}, \text{sk}_c, \text{nym}_c, \text{aux}_c, \text{cred}'_c, A'_c)$
31: Send (M, ψ) to PSD/Server

4 $\{0, 1\} \leftarrow \text{Server.Respond}(M, \psi)$

32: **if** $1 = \text{SGN.Verify}(\pi_c, \sigma_{\pi_c})$
33: **if** $0 = \text{VDF.Verify}(H(M), \tau, \ell, \psi)$
34: **return** $\perp$
35: **if** $1 = \text{CredVerify}(\text{PK}_{\text{FCC}}, \text{nym}_c, D_c; M)$ {AP-Assisted}
36: Parse $(m, \sigma_{\text{AP}}, \tau_{\text{AP}}) \leftarrow \Phi_c$
37: **if** $1 = \text{RLRS.Verify}(\mathcal{L}_{\text{ID}}, m, e_{\text{ID}}, \sigma_{\text{AP}}, \tau_{\text{AP}})$
38: PSD/Server **return** 1, and grant services
39: **elseif** $1 = \text{CredVerify}(\text{PK}_{\text{FCC}}, \text{nym}_c, A'_c)$ {ND-Assisted}
40: PSD/Server **return** 1, and grant services

games yields an adversary against one of the underlying hardness assumptions or against the binding properties of the event/context derivation mechanism.

Lemma 1 (Anonymous and Unlinkable Credential Authentication). *If an adversary $\mathcal{A}$ breaks credential unforgeability, anonymity, or unlinkability in SLAPX, then there exists a PPT adversary that either: (i) violates the unforgeability of the SPSEQ-UC-based credential/signing layer, (ii) distinguishes rerandomized, extended, or key-switched credential presentations from fresh ones contrary to the DDH-style indistinguishability assumptions used in the DAC privacy proof, or (iii) breaks the generic-group security guarantees on which the DAC instantiation relies.*

Proof of Lemma 1. We prove the claim via a sequence of hybrids. **Game G_0 .** This is the real DAC-backed authentication experiment in SLAPX, where $\mathcal{A}$ wins if it (i) produces an accepted credential presentation for unauthorized attributes, (ii) distinguishes the origin of a challenge presentation, or (iii) links two accepted presentations.

Game G_1 (privacy reduction). Replace honest credential showings with transcripts generated by the DAC challenger, preserving issuance and delegation behavior. Any non-negligible distinguishing advantage between G_0 and G_1 yields a PPT adversary against DAC presentation privacy (origin-hiding, derivation-privacy, or conversion-privacy). Hence, $|\Pr[G_0 = 1] - \Pr[G_1 = 1]| \leq \text{Adv}_{\text{DAC-anon}}^{\text{DAC}}(\lambda) + \text{Adv}_{\text{DAC-link}}^{\text{DAC}}(\lambda)$.

Game G_2 (unforgeability extraction). In G_1 , any accepted presentation revealing attributes not issued or delegated corresponds to a forgery against the DAC issuance/presentation layer. Thus, $\Pr[G_1 = 1] \leq \text{Adv}_{\text{DAC-uf}}^{\text{DAC}}(\lambda) + \text{negl}(\lambda)$. **Conclusion.** Combining the above games, $\text{Adv}_{\text{SLAPX}, \mathcal{A}}^{\text{auth}}(\lambda) \leq \text{Adv}_{\text{DAC-uf}}^{\text{DAC}}(\lambda) + \text{Adv}_{\text{DAC-anon}}^{\text{DAC}}(\lambda) + \text{Adv}_{\text{DAC-link}}^{\text{DAC}}(\lambda) + \text{negl}(\lambda)$. By the security of the SPSEQ-UC-based DAC construction, these advantages reduce to the unforgeability of the underlying signing layer and to DDH-style indistinguishability together with generic-group security assumptions used to prove origin-hiding, derivation-privacy, and conversion-privacy. $\square$

Lemma 2 (AP-Assisted PoL Authenticity and Event Binding). *If an adversary $\mathcal{A}$ produces a fresh accepted AP-assisted PoL object in SLAPX without honest AP endorsement, then there exists a PPT adversary that either: (i) breaks the DL-, DDL-, or q -SDH-based security of the instantiated RLRS scheme, or (ii) finds a collision or second preimage in the event/context-binding hash functions used to derive the message and event identifier.*

Proof of Lemma 2. Let $\mathcal{A}$ output a fresh accepted AP-assisted PoL object $\Phi^* = (m^*, \sigma^*, \tau^*)$ for context ctx^* without honest AP endorsement. **Game G_0 .** This is the real AP-assisted PoL experiment in SLAPX, where acceptance requires valid event/context binding and RLRS verification. **Game G_1 (event-binding consistency).** Recompute the event identifier e_{ID}^* and message m^* from the parsed context. Any change in acceptance between G_0 and G_1 implies that $\mathcal{A}$ produced two distinct inputs mapping to the same accepted binding or transformed a valid tuple under the same hash output. Hence, $|\Pr[G_0 = 1] - \Pr[G_1 = 1]| \leq \text{Adv}_{\text{Hash-coll}}^{\text{Hash}}(\lambda) + \text{Adv}_{\text{Hash-2pre}}^{\text{Hash}}(\lambda)$. **Game G_2 (RLRS extraction).** In G_1 , any accepted (σ^*, τ^*) on (m^*, e_{ID}^*) without honest AP participation yields a forgery against the RLRS scheme. Thus, $\Pr[G_1 = 1] \leq \text{Adv}_{\text{RLRS-uf}}^{\text{RLRS}}(\lambda) + \text{negl}(\lambda)$. **Conclusion.** $\text{Adv}_{\text{SLAPX}, \mathcal{A}}^{\text{ap-pol}}(\lambda) \leq \text{Adv}_{\text{RLRS-uf}}^{\text{RLRS}}(\lambda) + \text{Adv}_{\text{Hash-coll}}^{\text{Hash}}(\lambda) + \text{Adv}_{\text{Hash-2pre}}^{\text{Hash}}(\lambda) + \text{negl}(\lambda)$. By the security of the instantiated RLRS, $\text{Adv}_{\text{RLRS-uf}}^{\text{RLRS}}(\lambda)$ reduces to DL, DDL, and q -SDH hardness in the random oracle model. This proves the lemma. $\square$

Lemma 3 (AP-Assisted Replay and Event-Reuse Resistance). *If an adversary $\mathcal{A}$ causes two distinct accepted AP-assisted requests within the same event context to evade replay or reuse detection, then there exists a PPT adversary that either: (i) breaks the linkability/revocable-if-linked guarantees of the RLRS instantiation under DL, DDL, or q -SDH, or (ii) breaks collision or second-preimage resistance of the event-derivation hash.*

Proof of Lemma 3. Assume that $\mathcal{A}$ outputs two distinct accepted AP-assisted requests within the same event context while evading replay or reuse detection. **Game G_0 .** This is the real replay/reuse experiment in SLAPX, where the PSD verifies both PoL objects and then applies event-scoped replay and RLRS linkability checks. **Game G_1 (ideal event derivation).** Recompute the event identifier e_{ID} directly from the verified context and reject any inconsistency. Any change in acceptance from G_0 to G_1 implies a collision or second preimage in the event-derivation hash. Thus, $|\Pr[G_0 = 1] - \Pr[G_1 = 1]| \leq \text{Adv}_{\text{Hash-coll}}^{\text{Hash}}(\lambda) + \text{Adv}_{\text{Hash-2pre}}^{\text{Hash}}(\lambda)$. **Game G_2 (ideal linkability test).** Replace the concrete RLRS link procedure by an ideal same-signer detector. If $\mathcal{A}$ still causes two same-event accepted requests to evade detection, then it breaks the event-scoped linkability or revocable-if-linked guarantees of the RLRS construction. Hence, $\Pr[G_1 = 1] \leq \text{Adv}_{\text{RLRS-link}}^{\text{RLRS}}(\lambda) + \text{negl}(\lambda)$. **Conclusion.** $\text{Adv}_{\text{SLAPX}, \mathcal{A}}^{\text{reuse}}(\lambda) \leq \text{Adv}_{\text{RLRS-link}}^{\text{RLRS}}(\lambda) + \text{Adv}_{\text{Hash-coll}}^{\text{Hash}}(\lambda) + \text{Adv}_{\text{Hash-2pre}}^{\text{Hash}}(\lambda) + \text{negl}(\lambda)$.

For the instantiated RLRS, the linkability and revocable-if-linked guarantees reduce to DL, DDL, and q -SDH hardness in the random oracle model. $\square$

Lemma 4 (ND-Assisted Delegated PoL Soundness). *If an adversary $\mathcal{A}$ obtains an accepted ND-assisted delegated PoL in SLAPX without a valid nearby-device interaction, then there exists a PPT adversary that either: (i) breaks the underlying authenticated key-agreement assumption of the instantiated one-pass AKA protocol (e.g., GDH in the attached instantiation), (ii) succeeds against the distance-bounding layer beyond its stated fraud-resistance bound, (iii) forges the delegated credential chain contrary to the underlying DAC/SPSEQ-UC assumptions, or (iv) breaks hash-based context binding.*

Proof of Lemma 4. Suppose $\mathcal{A}$ outputs an accepted ND-assisted delegated PoL transcript (cred'_c, A'_c) without a valid nearby-device interaction. **Game G_0 .** This is the real ND-assisted experiment in SLAPX, where acceptance requires successful authenticated key establishment, distance-bounding acceptance, and a valid delegated credential chain. **Game G_1 (ideal session binding).** Replace the real one-pass AKA by an ideal session-key oracle. Any non-negligible change in acceptance implies a break of the underlying authenticated key-establishment security. Thus, $|\Pr[G_0 = 1] - \Pr[G_1 = 1]| \leq \text{Adv}^{\text{AKA}}(\lambda)$. **Game G_2 (ideal proximity acceptance).** Replace the concrete DBP by an ideal proximity predicate that accepts only honest nearby interactions. If $\mathcal{A}$ still succeeds, then it violates the DBP fraud-resistance bound. For the instantiated protocol, an out-of-range adversary succeeds with probability at most $(3/4)^n$ over n rapid rounds [22]. Hence, $|\Pr[G_1 = 1] - \Pr[G_2 = 1]| \leq (3/4)^n$. **Game G_3 (delegation extraction).** In G_2 , any accepted delegated extension A_c unsupported by a valid parent credential and delegation transcript yields either a forgery against the DAC delegation/signing layer or a break of hash-based context binding. Therefore, $\Pr[G_2 = 1] \leq \text{Adv}^{\text{DAC-uf}}(\lambda) + \text{Adv}^{\text{Hash-coll}}(\lambda) + \text{Adv}^{\text{Hash-2pre}}(\lambda) + \text{negl}(\lambda)$. **Conclusion.** $\text{Adv}^{\text{nd-pol}}_{\text{SLAPX}, \mathcal{A}}(\lambda) \leq \text{Adv}^{\text{AKA}}(\lambda) + (3/4)^n + \text{Adv}^{\text{DAC-uf}}(\lambda) + \text{Adv}^{\text{Hash-coll}}(\lambda) + \text{Adv}^{\text{Hash-2pre}}(\lambda) + \text{negl}(\lambda)$. For the instantiated ND-assisted branch, $\text{Adv}^{\text{AKA}}(\lambda)$ reduces to the corresponding Diffie-Hellman-type assumption (e.g., GDH), while the DBP term is bounded as above. $\square$

Lemma 5 (Authenticated Puzzle Issuance). *If an adversary $\mathcal{A}$ forges a fresh accepted puzzle token in SLAPX, then there exists a PPT adversary that breaks the EUF-CMA security of the instantiated puzzle-authentication signature scheme under its underlying hardness assumption.*

Proof of Lemma 5. Let $\mathcal{A}$ output a fresh accepted puzzle token-signature pair (π^*, σ^*) not issued by an honest PSD or server. **Game G_0 .** This is the real puzzle-issuance experiment in SLAPX, where each puzzle token is authenticated under the issuer's signing key. **Game G_1 (forgery extraction).** Record all puzzle messages signed by the honest issuer during the experiment. If $\mathcal{A}$ outputs a fresh accepted pair (π^*, σ^*) whose signed message was never issued before, then this pair is a valid EUF-CMA forgery against the underlying signature scheme. Hence, $\text{Adv}^{\text{puz-auth}}_{\text{SLAPX}, \mathcal{A}}(\lambda) \leq \text{Adv}^{\text{SIG-EUF}}(\lambda) + \text{negl}(\lambda)$. Thus, any successful forgery of an authenticated puzzle token contradicts the EUF-CMA security of the instantiated signature scheme under its underlying hardness assumption, proving the lemma. $\square$

Lemma 6 (Sequential Cost Enforcement and DoS Resilience). *If an adversary $\mathcal{A}$ causes accepted requests in SLAPX to bypass the framework's enforced per-request cost without incurring proportional work, then there exists a PPT adversary that either: (i) violates the repeated-squaring-based sequentiality assumption of the instantiated VDF, (ii) breaks the soundness of the VDF proof system in the chosen group of unknown order, (iii) forges authenticated puzzle material, (iv) or breaks the RLRS/hash-based replay and event-reuse enforcement.*

Proof of Lemma 6. Assume $\mathcal{A}$ causes accepted requests in SLAPX to bypass the enforced per-request cost without incurring proportional work. **Game G_0 .** This is the real DoS-resilience experiment where acceptance requires: (i) a valid authenticated puzzle token, (ii) a valid VDF solution and proof, and (iii) correct replay/rate-limiting enforcement in the AP-assisted branch. **Game G_1 (ideal puzzle authenticity).** Replace the puzzle-authentication mechanism by an ideal oracle. Any change in acceptance implies a forgery of the underlying signature scheme. Thus, $|\Pr[G_0 = 1] - \Pr[G_1 = 1]| \leq \text{Adv}^{\text{SIG-EUF}}(\lambda)$. **Game G_2 (ideal sequentiality).** Replace the VDF evaluation by an ideal delay oracle that enforces sequential work. If $\mathcal{A}$ still succeeds with reduced effort, then it violates the VDF sequentiality assumption. Hence, $|\Pr[G_1 = 1] - \Pr[G_2 = 1]| \leq \text{Adv}^{\text{VDF-seq}}(\lambda)$. **Game G_3 (ideal proof verification).** Replace the VDF proof verification by an ideal checker. Any remaining acceptance gap yields a break of VDF soundness: $|\Pr[G_2 = 1] - \Pr[G_3 = 1]| \leq \text{Adv}^{\text{VDF-snd}}(\lambda)$. **Game G_4 (ideal replay enforcement).** In the AP-assisted branch, replace replay/rate-limiting with an ideal event-scoped detector. If $\mathcal{A}$ still succeeds, it must bypass RLRS linkability or event binding. Thus, $\Pr[G_3 = 1] \leq \text{Adv}^{\text{RLRS-link}}(\lambda) + \text{Adv}^{\text{Hash-coll}}(\lambda) + \text{Adv}^{\text{Hash-2pre}}(\lambda) + \text{negl}(\lambda)$. **Conclusion.** $\text{Adv}^{\text{dos}}_{\text{SLAPX}, \mathcal{A}}(\lambda) \leq \text{Adv}^{\text{SIG-EUF}}(\lambda) + \text{Adv}^{\text{VDF-seq}}(\lambda) + \text{Adv}^{\text{VDF-snd}}(\lambda) + \text{Adv}^{\text{RLRS-link}}(\lambda) + \text{Adv}^{\text{Hash-coll}}(\lambda) + \text{Adv}^{\text{Hash-2pre}}(\lambda) + \text{negl}(\lambda)$. Under the concrete instantiation, VDF sequentiality follows from the repeated-squaring assumption in the chosen unknown-order group, while soundness follows from the Wesolowski construction [26]. This proves the lemma. $\square$

Theorem 1. *Under the security assumptions, the SLAPX framework satisfies the following security guarantees except with negligible probability in λ : (i) Anonymous and unlinkable UE authentication: no adversary can produce an accepted credential presentation for unauthorized attributes, distinguish which honest UE generated a challenge presentation, or link two accepted presentations beyond negligible advantage. More precisely, there exist PPT adversaries B_1, B_2, B_3 such that $\text{Adv}^{\text{auth}}_{\text{SLAPX}, \mathcal{A}}(\lambda) \leq \text{Adv}^{\text{DAC-uf}}_{B_1}(\lambda) + \text{Adv}^{\text{DAC-anon}}_{B_2}(\lambda) + \text{Adv}^{\text{DAC-link}}_{B_3}(\lambda) + \text{negl}(\lambda)$. (ii) AP-assisted PoL authenticity*

and event-scoped replay resistance: no adversary can produce a fresh accepted AP-assisted PoL without honest AP endorsement, nor reuse PoL objects within the same event context while evading the framework's RLRS-based replay and linkability checks. More precisely, there exist PPT adversaries B_4, B_5, B_6, B_7 such that $\text{Adv}^{\text{ap-pol}}_{\text{SLAPX}, \mathcal{A}}(\lambda) \leq \text{Adv}^{\text{RLRS-uf}}_{B_4}(\lambda) + \text{Adv}^{\text{RLRS-link}}_{B_5}(\lambda) + \text{Adv}^{\text{Hash-coll}}_{B_6}(\lambda) + \text{Adv}^{\text{Hash-2pre}}_{B_7}(\lambda) + \text{negl}(\lambda)$. (iii) ND-assisted delegated PoL soundness: no adversary can obtain an accepted delegated spatiotemporal credential extension without either breaking the authenticated key-establishment/distance-bounding subprotocols or forging a valid delegated credential chain. More precisely, there exist PPT adversaries B_8, B_9, B_{10}, B_{11} such that $\text{Adv}^{\text{nd-pol}}_{\text{SLAPX}, \mathcal{A}}(\lambda) \leq \text{Adv}^{\text{AKA}}_{B_8}(\lambda) + (3/4)^n + \text{Adv}^{\text{DAC-uf}}_{B_9}(\lambda) + \text{Adv}^{\text{Hash-coll}}_{B_{10}}(\lambda) + \text{Adv}^{\text{Hash-2pre}}_{B_{11}}(\lambda) + \text{negl}(\lambda)$. (iv) Authenticated and cost-enforcing DoS resilience: no adversary can cause accepted query or service requests to bypass the framework's per-request computational cost and authentication checks without either forging authenticated puzzle material, violating VDF sequentiality or soundness, or breaking event-scoped replay/rate-limiting enforcement. Precisely, there exist PPT adversaries $B_{12}, B_{13}, B_{14}, B_{15}, B_{16}$ such that $\text{Adv}^{\text{dos}}_{\text{SLAPX}, \mathcal{A}}(\lambda) \leq \text{Adv}^{\text{SIG-EUF}}_{B_{12}}(\lambda) + \text{Adv}^{\text{VDF-seq}}_{B_{13}}(\lambda) + \text{Adv}^{\text{VDF-snd}}_{B_{14}}(\lambda) + \text{Adv}^{\text{RLRS-link}}_{B_{15}}(\lambda) + \text{Adv}^{\text{Hash-coll}}_{B_{16}}(\lambda) + \text{Adv}^{\text{Hash-2pre}}_{B_{17}}(\lambda) + \text{negl}(\lambda)$.

Proof. We prove the theorem by reduction and case partitioning over the first security property violated by the adversary. Let $\mathcal{A}$ be any PPT adversary against SLAPX, and let $\text{Succ}_{\text{SLAPX}}(\mathcal{A})$ denote the event that $\mathcal{A}$ breaks at least one of the four claimed guarantees. We partition this event into the following disjoint cases: E_1 : $\mathcal{A}$ breaks anonymous and unlinkable UE authentication; E_2 : $\mathcal{A}$ breaks AP-assisted PoL authenticity or event-scoped replay resistance; E_3 : $\mathcal{A}$ breaks ND-assisted delegated PoL soundness; E_4 : $\mathcal{A}$ breaks authenticated and cost-enforcing DoS resilience. By the union bound, $\text{Adv}_{\text{SLAPX}, \mathcal{A}}(\lambda) = \Pr[\text{Succ}_{\text{SLAPX}}(\mathcal{A})] \leq$

$\sum_{i=1}^4 \Pr[E_i]$. For E_1 , Lemma 1 implies that any successful attack on credential-based authentication yields a PPT adversary against either DAC unforgeability, DAC anonymity, or DAC unlinkability. Hence, $\Pr[E_1] \leq Adv_{B_1}^{DAC-uf}(\lambda) + Adv_{B_2}^{DAC-anon}(\lambda) + Adv_{B_3}^{DAC-link}(\lambda) + negl(\lambda)$. For E_2 , Lemmas 2 and 3 show that any fresh forged AP-assisted PoL or any successful evasion of same-event replay detection yields a PPT adversary against RLRS unforgeability, RLRS linkability, or the collision/second-preimage resistance of the event-binding hash layer. Therefore, $\Pr[E_2] \leq Adv_{B_4}^{RLRS-uf}(\lambda) + Adv_{B_5}^{RLRS-link}(\lambda) + Adv_{B_6}^{Hash-coll}(\lambda) + Adv_{B_7}^{Hash-2pre}(\lambda) + negl(\lambda)$. For E_3 , Lemma 4 implies that any accepted ND-assisted delegated PoL produced without a valid nearby-device interaction yields a PPT adversary against the authenticated key-establishment/distance-bounding layer, the DAC delegation/issuance layer, or the hash-based context binding mechanism. Thus, $\Pr[E_3] \leq Adv_{B_8}^{AKA}(\lambda) + (3/4)^n + Adv_{B_9}^{DAC-uf}(\lambda) + Adv_{B_{10}}^{Hash-coll}(\lambda) + Adv_{B_{11}}^{Hash-2pre}(\lambda) + negl(\lambda)$. For E_4 , Lemmas 5 and 6 imply that any accepted request bypassing the enforced per-request cost must either forge authenticated puzzle material, violate VDF sequentiality or VDF soundness, or evade RLRS/hash-based replay and reuse checks. Hence, $\Pr[E_4] \leq Adv_{B_{12}}^{SIG-EUF}(\lambda) + Adv_{B_{13}}^{VDF-seq}(\lambda) + Adv_{B_{14}}^{VDF-snd}(\lambda) + Adv_{B_{15}}^{RLRS-link}(\lambda) + Adv_{B_{16}}^{Hash-coll}(\lambda) + negl(\lambda)$. Substituting the above bounds into the union bound yields exactly the four inequalities stated in the theorem. Under the stated security assumptions, each term on the right-hand side is negligible in λ , and therefore so is $Adv_{SLAP, A}(\lambda)$. $\square$

Data availability

No data was used for the research described in the article.

References

- [1] P. Agarwal, M. Manekia, T. Ahmad, A survey on citizens broadband radio service (cbrrs), *Electronics* 11 (23) (2022).
- [2] T. Chakraborty, S. Mitra, S. Mittal, Capow: Context-aware ai-assisted proof of work based ddos defense, 2023, ArXiv Preprint.
- [3] G. Gür, Expansive networks: Exploiting spectrum sharing for capacity boost and 6G vision, *J. Commun. Netw.* 22 (6) (2020) 444–454, <http://dx.doi.org/10.23919/JCN.2020.000037>.
- [4] M. Grissa, A.A. Yavuz, B. Hamdaoui, C. Tirupathi, Anonymous dynamic spectrum access and sharing mechanisms for the CBRS band, *IEEE Access* 9 (2021) 33860–33879.
- [5] A. Patil, S. Iyer, O.L. López, R.J. Pandya, K. Pai, A. Kalla, R. Kallimani, A comprehensive survey on spectrum sharing techniques for 5G/BSG intelligent wireless networks: Opportunities, challenges and future research directions, *Comput. Netw.* 253 (2024).
- [6] D.K. Jasim, S.B. Sadkhan, Cognitive radio network: Security and reliability trade-off-status, challenges, and future trend, in: 2021 1st Babylon International Conference on Information Technology and Science, BICITS, IEEE, 2021, pp. 149–153.
- [7] M. Grissa, A. Yavuz, B. Hamdaoui, An efficient technique for protecting location privacy of cooperative spectrum sensing users, in: 2016 IEEE Conference on Computer Communications Workshops, INFOCOM WKSHPS, IEEE, 2016, pp. 915–920.
- [8] J. Xin, M. Li, C. Luo, P. Li, Privacy-preserving spectrum query with location proofs in database-driven CRNs, in: 2016 IEEE Global Communications Conference, GLOBECOM, IEEE, 2016, pp. 1–6.
- [9] S. Darzi, A.A. Yavuz, Privacy-preserving and post-quantum counter denial of service framework for wireless networks, in: IEEE Military Communications Conference, MILCOM, IEEE, 2024, pp. 1076–1081.
- [10] S. Shi, Y. Xiao, W. Lou, C. Wang, X. Li, Y.T. Hou, J.H. Reed, Challenges and new directions in securing spectrum access systems, *IEEE Internet Things J.* 8 (8) (2021) 6498–6518.
- [11] Z. Newman, S. Servan-Schreiber, S. Devadas, Spectrum: High-bandwidth anonymous broadcast with malicious security, 2022.
- [12] N. Nguyen-Thanh, D.-T. Ta, V.-T. Nguyen, Spoofing attack and surveillance game in geo-location database driven spectrum sharing, *IET Commun.* 13 (1) (2019) 74–84.
- [13] G. Jakimoski, K. Subbalakshmi, Denial-of-service attacks on dynamic spectrum access networks, in: IEEE International Conference on Communications Workshops, IEEE, 2008, pp. 524–528.
- [14] S. Darzi, A.A. Yavuz, SLAP: Secure location-proof and anonymous privacy-preserving spectrum access, in: 2025 Silicon Valley Cybersecurity Conference, SVCC, IEEE, 2025, pp. 1–8.
- [15] M. Caleffi, A.S. Cacciapuoti, Database access strategy for TV white space cognitive radio networks, in: 2014 Eleventh Annual IEEE International Conference on Sensing, Communication, and Networking Workshops, SECON Workshops, IEEE, 2014, pp. 34–38.
- [16] S. Das, L. Zhu, J. Malyar, P. McCann, RFC 7545: Protocol to access white-space (PAWS) databases, 2015.
- [17] D. Order, Federal communications commission, 2023.
- [18] V. Chen, S. Das, L. Zhu, J. Malyar, P. McCann, Protocol to Access White-Space (Paws) Databases, Technical Report, 2015.
- [19] Y.-C. Liang, K.-C. Chen, G.Y. Li, P. Mahonen, Cognitive radio networking and communications: An overview, *IEEE Trans. Veh. Technol.* 60 (7) (2011) 3386–3407.
- [20] M. Thippeswamy, A.D. Prasanna, F. Takawira, Physical layer, data link layer, network layer, transport layer, and application layer in cognitive radio networks, in: Introduction To Cognitive Radio Networks and Applications, Chapman and Hall/CRC, 2016, pp. 187–200.
- [21] O. Mir, D. Slamanig, B. Bauer, R. Mayrhofer, Practical delegatable anonymous credentials from equivalence class signatures, *Proc. Priv. Enhancing Technol.* (2023).
- [22] H. Kilinc, S. Vaudenay, Efficient public-key distance bounding protocol, in: 22nd International Conference on the Theory and Application of Cryptology and Information Security, Springer, pp. 873–901.
- [23] S. Vaudenay, Private and secure public-key distance bounding: application to NFC payment, in: International Conference on Financial Cryptography and Data Security, Springer, 2015, pp. 207–216.
- [24] M.H. Au, J.K. Liu, W. Susilo, T.H. Yuen, Secure ID-based linkable and revocable-iff-linked ring signature with constant-size construction, *Theoret. Comput. Sci.* 469 (2013) 1–14.
- [25] D. Boneh, J. Boneau, B. Bünz, B. Fisch, Verifiable delay functions, in: Annual International Cryptology Conference, Springer, 2018, pp. 757–788.
- [26] B. Wesolowski, Efficient verifiable delay functions, *J. Cryptology* 33 (4) (2020) 2113–2147.
- [27] Z. Gao, H. Zhu, Y. Liu, M. Li, Z. Cao, Location privacy leaking from spectrum utilization information in database-driven cognitive radio network, in: Proceedings of the 2012 ACM Conference on Computer and Communications Security, 2012, pp. 1025–1027.
- [28] B. Bahrak, S. Bhattarai, A. Ullah, J.-M.J. Park, J. Reed, D. Gurney, Protecting the primary users' operational privacy in spectrum sharing, in: 2014 IEEE International Symposium on Dynamic Spectrum Access Networks, DYSPAN, IEEE, 2014, pp. 236–247.
- [29] M. Abuyaghi, S. Si-Mohammed, G. Shaker, C. Rosenberg, Positioning in 5g networks: Emerging techniques, use cases, and challenges, *IEEE Internet Things J.* (2024).
- [30] D. Lopez-Perez, A. Garcia-Rodriguez, L. Galati-Giordano, M. Kasslin, K. Doppler, IEEE 802.11 be extremely high throughput: The next generation of Wi-Fi technology beyond 802.11 ax, *IEEE Commun. Mag.* 57 (9) (2019) 113–119.
- [31] S. Rösler, N. Eising, A. Zubow, F. Dressler, Improving WiFi ranging through frequency diversity and mobility, in: 2025 IEEE 50th Conference on Local Computer Networks, LCN, IEEE, 2025, pp. 1–7.
- [32] G. Fuchsbaue, C. Hanser, D. Slamanig, Structure-preserving signatures on equivalence classes and constant-size anonymous credentials, *J. Cryptology* 32 (2019) 498–546.
- [33] E. Troja, S. Bakiras, Leveraging p2p interactions for efficient location privacy in database-driven dynamic spectrum access, in: 22nd ACM SIGSPATIA Conference, 2014, pp. 453–456.
- [34] Y. Li, L. Zhou, H. Zhu, L. Sun, Privacy-preserving location proof for securing large-scale database-driven cognitive radio networks, *IEEE Internet Things J.* 3 (4) (2015) 563–571.
- [35] M. Grissa, A.A. Yavuz, Location privacy in cognitive radios with multi-server private information retrieval, *IEEE Trans. Cogn. Commun. Netw.* (2019).
- [36] Y. Zeng, L. Xu, X. Yang, X. Yi, An efficient privacy-preserving protocol for database-driven cognitive radio networks, *Ad Hoc Netw.* 90 (2019) 101739.
- [37] NS3 network simulator, 2025, <https://www.nsnam.org/>. (Accessed January 2026).
- [38] N. Patricello, S. Lagen, B. Bojovic, L. Giupponi, An E2E simulator for 5G NR networks, *Simul. Model. Pr. Theory* 96 (2019) 101933.
- [39] Study on channel model for frequencies from 0.5 to 100 GHz, 2025, <https://portal.3gpp.org/desktopmodules/Specifications/SpecificationDetails.aspx?specificationId=3173>. (Accessed October 2025).
- [40] Y. Wang, C. Lin, Q.-L. Li, Y. Fang, A queueing analysis for the denial of service (DoS) attacks in computer networks, *Comput. Netw.* 51 (12) (2007) 3564–3573.
- [41] M. Duckett, J. Moisand, T. Anschutz, D. Kourkouzelis, P. Arberg, Accommodating a maximum transit unit/maximum receive unit (MTU/MRU) greater than 1492 in the point-to-point protocol over ethernet (pppoe), 2006, RFC 4638.

- [42] FCC, FCC regulations: 47 CFR 96.39, 2025, <https://www.ecfr.gov/current/title-47/chapter-I/subchapter-D/part-96/subpart-E/section-96.39>. (Accessed December 2025).
- [43] R. Zhu, L. Xu, Y. Zeng, X. Yi, Lightweight privacy preservation for securing large-scale database-driven cognitive radio networks with location verification, *Secur. Commun. Netw.* (2019).
- [44] M. Ul Hassan, M.H. Rehmani, M. Rehan, J. Chen, Differential privacy in cognitive radio networks: A comprehensive survey, *Cogn. Comput.* 14 (2) (2022) 475–510.
- [45] V. Bostanov, Client puzzle protocols as countermeasure against automated threats to web applications, *IEEE Access* 9 (2021).
- [46] I.M. Ali, M. Caprolu, R.D. Pietro, Foundations, properties, and security applications of puzzles: A survey, *ACM Comput. Surv.* 53 (4) (2020) 1–38.